Investments

A Quantitative Finance Master Guide

Mizuki Seika

From uncertain future payoffs to return forecasting, risk modeling, optimal portfolio allocation, implementation, and performance evaluation.

© 2026 Mizuki Seika. All original content is copyrighted by the author.

This work is licensed under the Creative Commons Attribution 4.0 International License (CC BY 4.0).

You are free to share and adapt this material, including for commercial purposes, provided that appropriate credit is given, a link to the license is included, and any modifications are indicated.

Author: Mizuki Seika

Contents

1 The Investment Problem: Future Benefits Under Uncertainty **6**
1.1 Present value and the return objective . . . 6
1.2 Expected versus realized returns . . . 8

2 Measuring Returns, Wealth Growth, and Risk **9**
2.1 Finite probability space and moments . . . 9
2.2 Simple returns, total returns, cumulative returns, and log returns . . . 10
2.3 Arithmetic vs geometric performance . . . 10
2.4 Empirical risk, higher moments, and long-run return evidence . . . 11
2.5 Share prices, stock splits, and equity values . . . 12
2.6 Portfolio return . . . 13

3 Economic Valuation: No-Arbitrage and the Stochastic Discount Factor **14**
3.1 Arbitrage, replication, and complete markets . . . 14
3.2 State prices and the stochastic discount factor . . . 15
3.3 Investor preferences and risk aversion . . . 16
3.4 Consumption-based equilibrium and the SDF . . . 17
3.5 Risk premia as covariance with the SDF . . . 18

4 The Risk-Neutral Branch: Pricing Derivatives **19**
4.1 Change of measure and risk-neutral pricing . . . 19
4.2 Options and the binomial example . . . 20
4.3 Black–Scholes under the physical and risk-neutral measures . . . 20
4.4 Delta hedging and Fundamental Theorem of Asset Pricing . . . 21

5 Where Expected Returns Come From: Valuation, Prediction, and Signals **22**
5.1 Dividend Discount Model (DDM) and Gordon growth . . . 22
5.2 Equilibrium cash yield and growth . . . 23
5.3 Equity-premium predictability . . . 24
5.4 OLS regression and slope significance . . . 24
5.5 In-sample vs out-of-sample R^2 . . . 25
5.6 Factor investing and characteristic forecasts . . . 25
5.7 Cross-sectional styles and return signals . . . 26

6 Measuring Joint Risk: Covariance, Diversification, and Risk Models **28**
6.1 Two risky assets: expected return, variance, covariance, correlation . . . 28
6.2 Diversification vs concentration . . . 29
6.3 CAPM as a risk model . . . 30
6.4 Characteristic, risk-factor, and pricing-factor distinctions . . . 30

7 Time-Varying Risk: Volatility Forecasting **32**
7.1 Conditional volatility models . . . 32

8 Portfolio Choice I: Choosing the Risky Composition **35**
8.1 Markowitz and asset allocation . . . 35
8.2 Portfolio frontier with risky assets . . . 35
8.3 N risky assets: matrix formulation and closed-form frontier . . . 36
8.4 Add the risk-free asset: CAL and tangency portfolio . . . 37

9 Portfolio Choice II: Choosing the Risk Scale **38**
9.1 Mean–variance preferences and indifference curves . . . 38
9.2 Risky-fund and risk-free composition . . . 39
9.3 Optimal risk exposure, leverage, and constraints . . . 39

10 Dynamic Asset Allocation **41**
10.1 Random walk and market timing . . . 41
10.2 Predictive market-timing implementation . . . 42
10.3 Detecting market timing with Treynor–Mazuy . . . 43

11 Equilibrium Asset Pricing: Portfolio Choice to CAPM **44**
11.1 CAPM assumptions and equilibrium market portfolio . . . 44
11.2 Deriving the capital asset pricing model . . . 45

12 Alpha, Beta, and Testing Asset-Pricing Models **47**
12.1 Testing CAPM: alpha and beta . . . 47
12.2 Alpha as mispricing and a Gordon-growth worked interpretation . . . 48
12.3 Why CAPM can fail . . . 49

13 Multifactor Asset Pricing **50**
13.1 SDF view of factor models . . . 50
13.2 Chen–Roll–Ross macro factors . . . 51

14 Fama–French and Modern Factor Investing **52**
14.1 Fama–French three-factor model . . . 52
14.2 Fama–French extensions / additional factors . . . 53

15 Long-Short Investing, Market Neutrality, and Economic Stories **54**
15.1 Investment-style economic stories . . . 54
15.2 Long-short and market-neutral investing . . . 54

16 Estimation Error and Robust Portfolio Construction **55**
16.1 Asset-class data, index funds, and implementation . . . 55
16.2 Estimation error: why theoretical optimal weights can fail . . . 56

17 Portfolio Implementation Constraints **58**
17.1 Portfolio constraints: turnover and max drawdown . . . 58

18 Active Management and Performance Evaluation **59**
18.1 Active portfolio management, tracking error, information ratio . . . 59
18.2 Buffett's Alpha . . . 60

19 Market Efficiency, Quantitative Research, and Limits of Prediction **61**
19.1 Machine learning: two cultures and finance-specific difficulties . . . 61
19.2 Efficient markets, near-efficient markets, and random walks . . . 61
19.3 Statistical failure modes and fund-manager evidence . . . 62
19.4 Empirical evidence on machine learning in asset markets . . . 62

A Complete Concept Index **64**
A.1 Uncertainty and measurement . . . 64
A.2 Economic valuation . . . 64
A.3 Derivative valuation . . . 65
A.4 Conditional expected returns . . . 65
A.5 Joint and conditional risk . . . 65
A.6 Static portfolio construction . . . 66
A.7 Dynamic portfolio allocation . . . 66
A.8 Equilibrium and testing . . . 66
A.9 Trading and robustness . . . 67
A.10 Evaluation and market structure . . . 67

B Complete Formula Index **68**
B.1 Returns, compounding, and distributional moments . . . 68
B.2 Cash-flow valuation and predictive regressions . . . 68
B.3 State prices, stochastic discount factors, and risk-neutral pricing . . . 69
B.4 Black–Scholes pricing and delta . . . 70
B.5 Static and dynamic asset allocation . . . 70
B.6 Volatility and conditional variance models . . . 71
B.7 Mean–variance portfolio optimization . . . 71
B.8 CAPM and multifactor expected returns . . . 72
B.9 Additional identities and implementation equations . . . 72

C Final Investment Discipline Map **74**
C.1 Two connected pricing problems . . . 74

D Quantitative Investment Research Workflow **76**
D.1 Operational interpretation . . . 76

PART 1

The Investment Problem: Future Benefits Under Uncertainty

The object of investment is a dated, state-contingent payoff. Capital is committed at date zero, while the payoff is delivered in a future economic state that is not known today. Two decisions are therefore inseparable: determining the economic price of uncertainty, and determining the amount of wealth to expose to it.

$$P_0 = \frac{X_T}{(1+R)^T} \qquad \text{(deterministic benchmark).}$$

For a risky payoff, discounting the mean payoff by an arbitrary quoted rate is not a general pricing principle: investors attach different values to cash flows realized in different states. Later, the stochastic discount factor resolves precisely this difficulty.

A realized return is an outcome; an expected return is a conditional forecast. Writing

$$R_{t+1} = E_t[R_{t+1}] + \varepsilon_{t+1}, \qquad E_t[\varepsilon_{t+1}] = 0,$$

separates the researcher's forecast from the surprise. Investment decisions must depend on information available when positions are taken, not on the surprise known afterward.

A useful local representation of risk preferences is

$$U \simeq E[R] - \frac{\gamma}{2}\,\mathrm{Var}(R), \qquad \gamma > 0.$$

The risk penalty explains why a higher mean is not always worth additional exposure. This guide develops the economic value of payoffs, estimates their expected returns and covariances, and then chooses *which* risky assets to hold and *how much* total risk to take.

1.1 Present value and the return objective

Investment means paying something today to receive cash flows later. Discounting translates a future cash flow into dollars today.

$$P_0 = \frac{X_T}{(1+R)^T}$$

- X_T = cash flow received at date T; R = return/discount rate associated with the cash-flow stream.

$$R = \frac{\text{what you get}}{\text{what you paid}} - 1$$

- A riskless investment has a non-random rate of return; these rates are typically called interest rates.
- A risky investment has a random return. The object of interest ex ante is E[R], not the one realized outcome.

$$\mathbb{E}[R] = \frac{\mathbb{E}[\text{what you get}]}{\text{what you paid}} - 1$$

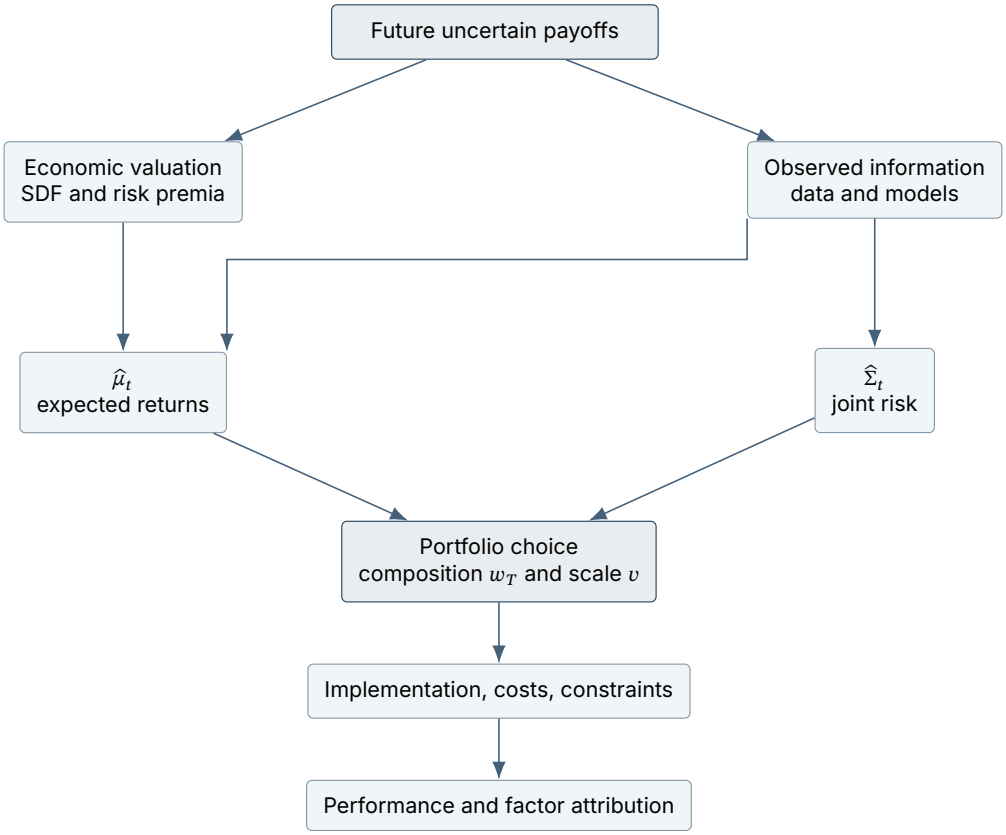


Figure 1: The investment decision as a sequence from economic uncertainty to measurable positions.

1.2 Expected versus realized returns

$$R = \mathbb{E}[R] + \varepsilon$$

• Expected component: the systematic/predictable component inferred from historical data, models, or other forecasts given current information.

• Unexpected component ε: shock from unforeseen news/events; random/idiosyncratic part of realized return.

Expected-return model families listed in the underlying analysis:

• CAPM: E[R] = R_f + β(E[R_M] − R_f).

• Fama–French models and broader factor models.

• Statistical models such as principal components analysis (PCA).

PART 2

Measuring Returns, Wealth Growth, and Risk

The fundamental observable is a sequence of wealth changes, not a raw price series. Return definitions must specify the treatment of cash distributions, the compounding convention, and the measurement horizon. A statistical risk measure must specify whether it concerns the full return distribution or only losses below a threshold.

$$R_t^{\text{total}} = \frac{P_t - P_{t-1} + D_t}{P_{t-1}}, \qquad r_t = \log(1 + R_t),$$

$$1 + R_{0:T} = \prod_{t=1}^{T} (1 + R_t).$$

The arithmetic mean estimates expected one-period performance under stationarity. The geometric mean describes realized compound wealth growth. Their difference is not a technicality: volatility makes repeated proportional gains and losses asymmetric in terminal wealth. Variance and standard deviation describe dispersion; skewness and kurtosis describe shape and tails.

2.1 Finite probability space and moments

• A finite probability space has a nonempty finite state space Ω and probability measure P with P(ω)∈[0,1] and Σω P(ω)=1.

• A random variable X is a real-valued function on Ω.

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} P(\omega) X(\omega)$$

$$\mathbb{E}[R] = \sum P(\omega) R(\omega)$$

$$V(R) = \mathbb{E}[(R - \mathbb{E}[R])^2]$$

$$V_d(R) = \mathbb{E}[\min(R - \tau, 0)^2]$$

$$\sigma(R) = \sqrt{V(R)} \quad \text{and} \quad \sigma_d(R) = \sqrt{V_d(R)}$$

$$\text{Skew}(R) = \frac{\mathbb{E}[(R - \mathbb{E}[R])^3]}{\sigma^3}$$

$$\text{Kurt}(R) = \frac{\mathbb{E}[(R - \mathbb{E}[R])^4]}{\sigma^4},$$

$$\text{excess kurtosis} = \text{kurtosis} - 3$$

Interpretation: Variance treats upside and downside symmetrically; downside variance only penalizes returns below threshold τ. Skewness captures asymmetry; kurtosis captures tail thickness relative to the scale of volatility.

2.2 Simple returns, total returns, cumulative returns, and log returns

Price (simple) return

$$R_t^x = \frac{P_t}{P_{t-1}} - 1$$

Total (simple) return

$$R_t = \frac{P_t + D_t}{P_{t-1}} - 1$$

Dividend-yield identity as written on the illustration

$$\frac{D_t}{P_t} = \frac{1 + R_t}{1 + R_t^x} - 1$$

Timing/indexing conventions matter for dividend-yield identities; the formula above preserves the illustration's notation.

Cumulative simple price return

$$R_{t\to T}^x = \prod_{s=t}^{T}(1 + R_s^x) - 1$$

Cumulative simple total return

$$R_{t\to T} = \prod_{s=t}^{T}(1 + R_s) - 1$$

Log price return

$$r_t^x = \ln(1 + R_t^x); \text{log total return} : r_t = \ln(1 + R_t)$$

Cumulative log return

$$r_{t\to T} = \sum_{s=t}^{T} r_s$$

KEY: Simple returns compound multiplicatively; log returns aggregate additively.

2.3 Arithmetic vs geometric performance

Arithmetic average simple return

$$\bar{R} = \frac{1}{T}\sum_t R_t$$

Arithmetic average log return

$$\bar{r} = \frac{1}{T}\sum_t r_t$$

Geometric average simple return

$$\bar{R}_G = \left[\prod_t (1 + R_t)\right]^{\frac{1}{T}} - 1$$

- Arithmetic and geometric averages can be computed for price returns or total returns.
- The arithmetic average of simple returns ignores compounding/volatility drag. Log returns are additive; the arithmetic mean of log returns maps to compound growth.

Quarter	Price	Simple return	Gross return
0	100	—	—
1	108	8.0%	1.080
2	104	−3.7%	0.963
3	110	5.8%	1.058
4	115	4.5%	1.045

Numerical example: arithmetic quarterly average ≈3.7%; multiplying by 4 gives ≈14.6%. Geometric quarterly average ≈3.6%; compounding for four quarters gives ≈15.0%.

Annual return from 12 monthly returns

$$R_a = \prod_{m=1}^{12} (1 + R_m) - 1$$

2.4 Empirical risk, higher moments, and long-run return evidence

Sample variance

$$V(R) = \frac{1}{T-1} \sum_t (R_t - \bar{R})^2$$

Sample downside variance

$$V_d(R) = \frac{1}{T-1} \sum_t \min(R_t - \tau, 0)^2$$

$$\sigma = \sqrt{V}; \sigma_d = \sqrt{V_d}$$

$$\text{Skew}_{sample} = \frac{1}{\sigma^3 T} \sum_t (R_t - \bar{R})^3$$

$$\text{Kurt}_{\text{sample}} = \frac{1}{\sigma^4 T} \sum_t (R_t - \bar{R})^4,$$

$$\text{excess kurtosis} = \text{Kurt}_{\text{sample}} - 3$$

Long-run S&P 500 evidence (Robert Shiller, 1871–present)

Statistic	Price return	Total return
Observations	1865	1865
Arithmetic mean	5.77%	9.98%
Geometric mean	4.89%	9.39%
Std. dev.	14.02%	14.01%
Downside risk	8.52%	8.14%
Skewness	0.3744	0.4405
Excess kurtosis	16.7559	17.4493

Evidence: Dividends materially increase long-run performance; arithmetic means exceed geometric means; return distributions display extremely high excess kurtosis (fat tails). The monthly-return histogram is sharply centered near zero with large tail events.

Additional historical-equity evidence

An October 2025 vintage of the Robert Shiller S&P 500 data yields a separate 1,840-observation snapshot. Its historical estimates should not be mixed with the later 1,865-observation sample above.

Statistic	Price return	Total return
Observations	1840	1840
Annual mean	5.61%	9.86%
Annual std. dev.	14.05%	14.04%
Monthly min.	−26.47%	−26.19%
Monthly max.	50.30%	51.31%

2.5 Share prices, stock splits, and equity values

The Tesla charts compare raw share price, cumulative split factor, and split-adjusted share price. Raw prices can jump mechanically around splits; adjusted prices are needed for economically meaningful return comparisons.

$$\text{ME} = \text{market price per share} \times \text{shares outstanding}$$

- ME cannot be negative and can differ greatly from accounting book value.

$$\text{BE} = \text{book value of assets} - \text{book value of liabilities}$$

- BE can be negative.

Market-to-book (price-to-book) ratio

$$MB = \frac{ME}{BE}$$

- Low MB → “value” stock; high MB → “growth” stock.

Stock split: A split increases shares outstanding and reduces the price per share proportionally, so total market capitalization is unchanged. Example: a 2-for-1 split doubles shares and halves the share price; economic ownership is unchanged.

The SPY vs TSLA rate-of-return chart reinforces that returns—not raw dollar prices—are the comparable performance object.

2.6 Portfolio return

Two-asset portfolio

$$R_{p,t} = wR_{1,t} + (1 - w)R_{2,t}$$

Portfolio returns are linear in weights, while portfolio variance depends on cross-asset covariance.

PART 3

Economic Valuation: No-Arbitrage and the Stochastic Discount Factor

What is the value today of one unit paid in a particular future state? No-arbitrage begins with an operational answer: two portfolios delivering identical state-by-state cash flows must have the same price. In a complete finite-state market, all payoffs can be replicated from state-contingent claims.

$$P(X) = \sum_{\omega} \zeta(\omega) X(\omega), \qquad \zeta(\omega) = P(\omega) M(\omega).$$

Here $P(\omega)$ is a physical probability; $\zeta(\omega)$ is a *dollar price per future state-contingent dollar*; $M(\omega)$ is a random discount rate expressed as a state-dependent payoff multiplier. The distinction is necessary because a rare dollar paid during a recession may be more valuable than an equally likely dollar paid during prosperity.

For an interior consumption choice, the marginal-utility trade-off between today and tomorrow gives

$$M_{t+1} = \beta \frac{u'(C_{t+1})}{u'(C_t)}, \qquad P_t(X_{t+1}) = E_t[M_{t+1} X_{t+1}].$$

Concave utility makes marginal utility high when consumption is scarce. An asset delivering less in high-M states is especially painful to hold; it must compensate investors with a positive expected premium. Formally, since $E[M] = (1 + R_f)^{-1}$,

$$1 = E[M(1 + R_i)], \qquad \boxed{E[R_i - R_f] = -(1 + R_f) \operatorname{Cov}(M, R_i - R_f)}.$$

Thus standalone variance is not the fundamental determinant of compensation. The relevant quantity is covariance with economically bad states.

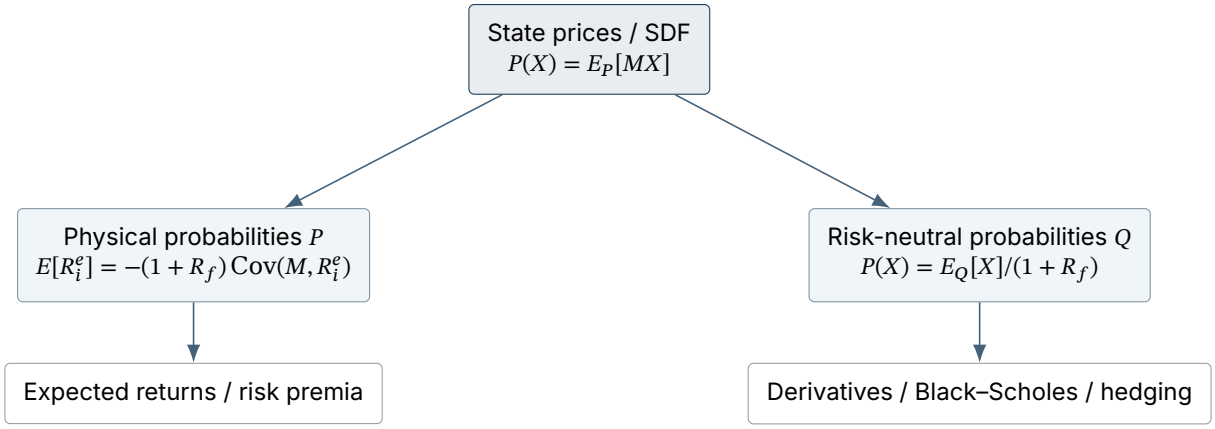


Figure 2: The same SDF supports both equilibrium premia and arbitrage-free derivative pricing.

3.1 Arbitrage, replication, and complete markets

• Arbitrage opportunity: a portfolio/trade with zero cost today (or a negative price, so you receive money today), zero probability of any future loss, and positive probability of a positive future payoff.

• Law of one price (LOP): two assets with identical payoffs in every state must have the same price; otherwise an arbitrage exists.

Loctite quantity pricing, the 3Com/Palm negative-stub case, SLABS, and the Google bond offer examples of how no-arbitrage relations can become stressed in real markets when frictions or limits to arbitrage prevent immediate convergence. Their role is conceptual: LOP is the benchmark, not a claim that every observed market is frictionless.

Complete markets and Arrow–Debreu securities

• Markets are complete if for every state ω there exists a state-contingent claim paying \$1 in ω and \$0 in all other states.

• Its price ζ(ω) is the state price / Arrow–Debreu price. ζ(ω)>0 in an arbitrage-free market.

• High state price corresponds to a "bad" state: one dollar delivered there is especially valuable; low state price corresponds to a "good" state.

Claim price

$$P(X) = \sum_{\omega \in \Omega} \zeta(\omega) X(\omega)$$

Risk-free bond (payoff 1 in every state)

$$P_f = \sum_{\omega} \zeta(\omega) = \frac{1}{1 + R_f}$$

3.2 State prices and the stochastic discount factor

$$M(\omega) = \frac{\zeta(\omega)}{P(\omega)} > 0$$

$$P(X) = \sum P(\omega) M(\omega) X(\omega) = \mathbb{E}[MX]$$

$$\mathbb{E}[M] = \frac{1}{1 + R_f}$$

• M is the ratio of state price to physical probability. High M corresponds to bad states because marginal dollars are particularly valuable there.

• The CAPM later imposes an SDF that is a decreasing function of market excess return: market booms are low-M states; market crashes are high-M states.

Worked 2-state state-price example

Observation inputs: two equally likely states, ζ_1=\$0.35, ζ_2=\$0.60; a risky payoff is \$120 in state 1 and \$50 in state 2.

$$M_1 = \frac{0.35}{0.50} = 0.70\,; \quad M_2 = \frac{0.60}{0.50} = 1.20$$

$$P(X) = 0.35 \times 120 + 0.60 \times 50 = \$72$$

$$P_f = 0.35 + 0.60 = 0.95 \;\Rightarrow\; R_f = \frac{1}{0.95} - 1 \approx 5.263\%$$

$$\mathbb{E}_P[X] = 0.5 \times 120 + 0.5 \times 50 = 85 \;\Rightarrow\; \mathbb{E}_P[R] = \frac{85}{72} - 1 \approx 18.056\%$$

$$\text{Expected excess return} \approx 18.056\% - 5.263\% = 12.79\%$$

State-price matrix representation

With S states and S linearly independent traded claims: P is the S×1 price vector, X is an S×S payoff matrix with full rank, and Z is the S×1 vector of state prices.

$$P = XZ \Rightarrow Z = X^{-1}P$$

3.3 Investor preferences and risk aversion

• Utility maps outcomes into a numerical ranking. The cardinal number itself has no direct economic meaning; ordering is what matters.

• Non-satiation: utility rises with wealth/consumption. Concavity: marginal utility falls as wealth rises.

Power utility

$$u(C) = \frac{C^{1-\gamma}}{1-\gamma}$$

$$\text{Special case } \gamma = 1: \quad u(C) = \ln C$$

• γ>0 risk-averse; γ=0 risk-neutral (linear utility); γ<0 risk-loving.

The analysis contrasts very risk-averse investors, whose utility changes sharply when moving from rich to poor states, with risk-tolerant investors, whose utility is flatter.

Jensen's inequality and risk aversion

Concavity

$$\lambda u(a) + (1-\lambda)u(b) \leq u(\lambda a + (1-\lambda)b)$$

If u is concave

$$\mathbb{E}[u(X)] \leq u(\mathbb{E}[X])$$

For a zero-mean gamble X

$$\mathbb{E}[u(C+X)] \leq u(C)$$

KEY: Risk aversion means the expected utility of a fair gamble is below the utility of receiving its expected payoff with certainty.

Certainty-equivalent price of a gamble

For a 50/50 gamble paying $500 or $1,000, the price C that makes the investor indifferent solves E[u(X)] = u(C). The evidence reports:

γ	**Price**
0	\$750
1	\$707
2	\$667
3	\$632
4	\$606
5	\$586
10	\$540
20	\$519
50	\$507

As γ rises, risk aversion rises and the certainty equivalent falls toward the low payoff.

3.4 Consumption-based equilibrium and the SDF

Investor chooses current consumption C_0 and state-dependent future consumption $C(\omega)$:

$$\max\ u(C_0) + \beta \sum_{\omega} P(\omega) u(C(\omega)), \qquad \beta < 1$$

Budget

$$C_0 + \sum_{\omega} \zeta(\omega) C(\omega) = W_0$$

β captures the time value of money. The first-order condition implies:

$$\zeta(\omega) = \frac{\beta P(\omega) u'(C(\omega))}{u'(C_0)}$$

$$M(\omega) = \frac{\zeta(\omega)}{P(\omega)} = \frac{\beta u'(C(\omega))}{u'(C_0)}$$

KEY: The SDF is high when future marginal utility is high—i.e., in bad consumption states. Assets that pay in those states are valuable hedges and can offer lower expected returns.

Worked power-utility consumption example

Observation inputs: good-state consumption +5%, bad-state −5%; equal probabilities; γ=2; β=0.95.

$$\text{For power utility, } u'(C) \propto C^{-\gamma}; \text{normalize } C_0 = 1.$$

$$M_good = 0.95 \times (1.05)^{-2} \approx 0.86168$$

$$M_bad = 0.95 \times (0.95)^{-2} \approx 1.05263$$

$$\zeta_good = 0.5 M_good \approx 0.43084; \zeta_bad \approx 0.52632$$

$$P_f = \zeta_good + \zeta_bad \approx 0.95716 \Rightarrow R_f \approx 4.48\%$$

3.5 Risk premia as covariance with the SDF

For an asset with positive price P and stochastic payoff X:

$$1 + R(\omega) = \frac{X(\omega)}{P}$$

Fundamental pricing equation

$$1 = \mathbb{E}[M(1 + R)]$$

Observation's expected-gross-return form

$$\mathbb{E}[1 + R] = (1 + R_f)(1 - \mathrm{Cov}(M, R))$$

$$\text{Excess return } R^e = R - R_f : \quad \mathbb{E}[R^e] = -(1 + R_f)\,\mathrm{Cov}(M, R^e)$$

KEY: A positive risk premium requires negative covariance with M: the asset tends to perform poorly precisely when marginal utility is high (bad times), so investors demand compensation.

CAPM restriction introduced here: M(ω)=a+bR_m^e(ω), where R_m^e is market excess return.

PART 4

The Risk-Neutral Branch: Pricing Derivatives

The SDF has a second interpretation: it reweights physical probabilities into pricing probabilities. Define

$$Q(\omega) = (1 + R_f)P(\omega)M(\omega).$$

Because $E_P[M] = (1 + R_f)^{-1}$, the new probabilities sum to one, and

$$\boxed{P_0(X) = E_P[MX] = \frac{E_Q[X]}{1 + R_f}}.$$

The measure Q is a valuation device, not a prediction of actual future frequencies. It absorbs risk prices into probability weights so that traded discounted prices become martingales under the usual no-arbitrage assumptions. In derivative models the physical drift μ is replaced by the risk-free drift; exposure to economic risk is still present in the state reweighting and in option prices.

4.1 Change of measure and risk-neutral pricing

Let P be the physical probability measure and P~ another strictly positive probability measure. Define likelihood ratio:

$$Z(\omega) = \frac{\widetilde{P}(\omega)}{P(\omega)}; \quad \mathbb{E}[Z] = 1; \quad \widetilde{\mathbb{E}}[X] = \mathbb{E}[ZX]$$

Choose the new probability measure so the SDF is absorbed into probabilities:

$$\frac{\widetilde{P}(\omega)}{P(\omega)} = \frac{M(\omega)}{\mathbb{E}[M]}$$

$$P(X) = \frac{1}{1 + R_f}\widetilde{\mathbb{E}}[X]$$

- P~ is the risk-neutral/risk-adjusted measure, conventionally denoted Q.

Under Q

$$\widetilde{\mathbb{E}}[R] = R_f$$

> **KEY: Risk-neutral does not mean real investors are risk neutral. Q is a reweighted probability measure that embeds risk adjustment, letting us discount Q-expected payoffs at the risk-free rate.**

Monte Carlo valuation under Q

$$\widetilde{\mathbb{E}}[X] \approx \frac{1}{N}\sum_{i=1}^{N} X_i$$

- Strong Law of Large Numbers: the simulation average converges to the Q-expectation as N grows.
- Advantages listed: handles complex payoffs, multiple underlyings, jumps, and stochastic volatility.

Three equivalent complete-market pricing representations

Representation	Pricing equation
State prices	P(X)=Σζ(ω)X(ω)
SDF	P(X)=E[MX]
Risk-neutral	P(X)=E_Q[X]/(1+R_f)

4.2 Options and the binomial example

• A call gives the buyer the right, not the obligation, to buy the underlying at strike K on expiration day.

Call payoff

$$X(\omega) = \max\{S(\omega) - K, 0\}$$

Call price

$$C = \mathbb{E}[M \cdot \max(S - K, 0)] = \frac{\mathbb{E}_Q[\max(S - K, 0)]}{1 + R_f}$$

Worked 2-state call option

Using ζ_1=.35, ζ_2=.60 and P_f=.95: risk-neutral probabilities are proportional to state prices.

$$q_1 = \frac{0.35}{0.95} \approx 0.368421\ ; \quad q_2 = \frac{0.60}{0.95} \approx 0.631579$$

Underlying is 120 or 50; K=90, so call payoffs are 30 and 0.

$$C = 0.35 \times 30 + 0.60 \times 0 = \$10.50$$

4.3 Black–Scholes under the physical and risk-neutral measures

• Time is continuous and measured in years; expiration payoff X_T=max(S_T−K,0).

Under P

$$\ln\left(\frac{S_T}{S_0}\right) \sim N\left(\left(\mu - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right)$$

$$C_0 = e^{-r_f T}\mathbb{E}_Q[X_T] \quad (r_f \text{ is the continuously compounded/log risk-free rate})$$

Girsanov change of probability measure

$$Z_T = \frac{dQ}{dP} = \exp\left(-\frac{\theta^2 T}{2} - \theta\varepsilon_T\right),$$

$$\theta = \frac{\mu - r_f}{\sigma}, \quad \varepsilon_T \sim N(0, T)$$

Under Q

$$\ln\left(\frac{S_T}{S_0}\right) \sim N\left(\left(r_f - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right)$$

The derivation decomposes the call payoff with the indicator 1{S_T>K} and uses a truncated-normal exponential-moment lemma.

European call formula

$$C_0 = S_0\Phi(d_1) - Ke^{-r_f T}\Phi(d_2)$$

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r_f + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

4.4 Delta hedging and Fundamental Theorem of Asset Pricing

Call delta

$$\Delta = \frac{\partial C_0}{\partial S_0} = \Phi(d_1)$$

- Delta-hedged portfolio on the illustration: long 1 option + short Δ shares; Π=V−ΔS.
- It is instantaneously risk-free in the continuous-time model but requires continuous rebalancing because Δ changes with S and time.

Fundamental Theorem of Asset Pricing (analysis statement): in complete and arbitrage-free markets, there exists a risk-neutral measure under which all assets, when priced relative to a numeraire, are martingales.

Martingale

$$\mathbb{E}_Q[X_{t+1}|F_t] = X_t$$

Non-dividend stock

$$S_t = \mathbb{E}_Q\left[\frac{S_{t+1}}{1+R_f}\middle| F_t\right]$$

PART 5

Where Expected Returns Come From: Valuation, Prediction, and Signals

Portfolio optimization is conditional on an estimated expected-return vector $\mu_t = E_t[R_{t+1}]$. Since future returns are hidden, forecasting becomes a separate empirical problem. Three routes should not be confused: *fundamental valuation* links prices to cash flows and discount rates; *predictive regression* estimates conditional time-series relationships; *characteristic models* exploit differences across securities.

$$\text{Cash yield + growth} \longrightarrow \widehat{\mu}_t, \qquad R^e_{m,t+1} = a + bF_t + \varepsilon_{t+1}, \qquad \widehat{\mu}^e_{i,t} = \delta_0 + \sum_j \delta_j z_{j,i,t}.$$

A dividend yield can be high because expected returns are high, expected growth is weak, or the market has changed its beliefs. Statistical significance in a fitted sample does not automatically establish a usable forecast. All information must be lagged to the investment decision time and validated out of sample.

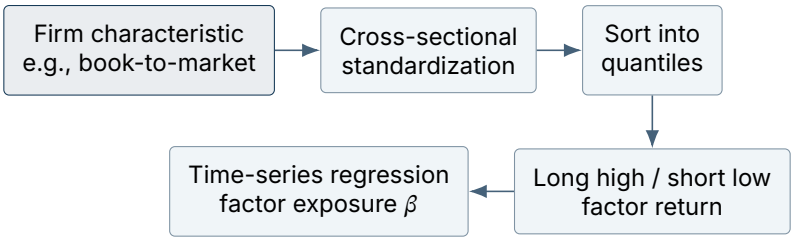


Figure 3: A characteristic, a traded factor portfolio, and a factor beta are different statistical objects.

5.1 Dividend Discount Model (DDM) and Gordon growth

Gross return

$$1 + R_{t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}$$

One-step pricing relation

$$P_t = \mathbb{E}_t\left[\frac{P_{t+1} + D_{t+1}}{1 + R_t}\right]$$

With constant expected return R, iterating S periods expresses price as the present value of dividends through t+S plus the discounted terminal price.

$$P_t = \mathbb{E}_t\left[\sum_{s=1}^{S}\left(\frac{1}{1+R}\right)^s D_{t+s}\right] + \mathbb{E}_t\left[\left(\frac{1}{1+R}\right)^S P_{t+S}\right]$$

As S→∞ and the discounted terminal-price term vanishes, price is the present value of all future dividends.

$$P_t = \mathbb{E}_t\left[\sum_{s=1}^{\infty}\left(\frac{1}{1+R}\right)^s D_{t+s}\right]$$

Constant dividend growth

$$\mathbb{E}_t[D_{t+s}] = (1+G)^{s-1}\mathbb{E}_t[D_{t+1}]$$

Gordon growth model

$$P_t = \frac{\mathbb{E}_t[D_{t+1}]}{R-G} \quad \text{(finite when } R > G\text{)}$$

Price-dividend ratio

$$\frac{P_t}{\mathbb{E}_t[D_{t+1}]} = \frac{1}{R-G}$$

Expected return

$$R = \frac{\mathbb{E}_t[D_{t+1}]}{P_t} + G$$

KEY: A valuation ratio can move because expected cash-flow growth changes, the discount rate changes, or both. A high price-dividend ratio is not mechanically "good" or "bad" without separating growth and discount-rate news.

Historical charts: dividend yield varies substantially through time and spikes in stressed periods; price-dividend and CAPE valuation ratios also vary strongly and have risen to high levels in some modern episodes.

5.2 Equilibrium cash yield and growth

$$\frac{D}{P} = R - G$$

$$G = \left(1 - \frac{D}{E}\right) \times ROE$$

D/E is payout ratio. When ROE equals the expected return in long-run equilibrium, the illustration derives:

$$R = \frac{E}{P}$$

KEY: In the stated long-run equilibrium, expected return equals the earnings yield.

5.3 Equity-premium predictability

Market excess return

$$R^e_{m,t} = R_{m,t} - R_{f,t}$$

Predictive regression

$$R^e_{m,t} = a + bF_{t-1} + \varepsilon_t$$

Prominent lagged predictors in the underlying analysis: dividend-price ratio; Shiller price-earnings/CAPE ratio; term spread; credit/default spread. Additional predictors: bond yield spreads, corporate spreads, inflation, investment rate, equity issuance, consumption-wealth ratio, option variance risk premium, and book-to-market for the cross-section.

- In-sample fit can look good because the model is estimated on the same data used to evaluate it.
- Overfitting produces patterns that do not persist out of sample.
- Tools listed to combat/assess this: rolling regressions, LASSO, ridge, neural nets, and explicit out-of-sample evaluation.

5.4 OLS regression and slope significance

Linear model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

OLS chooses β_0, β_1 to minimize the sum of squared residuals e_i = y_i − ŷ_i.

$$min_{\hat\beta_0,\hat\beta_1} \sum_i (y_i - \hat\beta_0 - \hat\beta_1 x_i)^2$$

$$\hat\beta_1 = \frac{\sum (x_i - \bar x)(y_i - \bar y)}{\sum (x_i - \bar x)^2} = \frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)}$$

$$\hat\beta_0 = \bar y - \hat\beta_1 \bar x$$

t-test for β_1

- H_0: $\beta_1=0$ (no linear relationship); H_1: $\beta_1\neq 0$ (two-sided relationship).

$$t = \frac{\hat\beta_1}{\mathrm{SE}(\hat\beta_1)} \quad \text{with df} \quad n-2 \quad \text{under } H_0 \text{ and classical assumptions}$$

$$\mathrm{SE}(\hat\beta_1) = \sqrt{\frac{\mathrm{SSR}}{(n-2)\sum_i (x_i - \bar x)^2}}$$

- Critical-value method: reject H_0 if |t| > t_{α/2,n−2}; for a large sample at α=5%, approximate critical values are ±1.96.
- α = probability of a Type I error: rejecting a true null.
- p-value = probability, assuming H_0, of seeing a test statistic at least as extreme as the observed one. Reject when p<α.

```
import statsmodels.api as sm

reg = sm.OLS(y, sm.add_constant(X)).fit()

print(reg.summary())
```

Return-predictability evidence in Cochrane (2011)

Horizon k	Slope b	t(b)	R^2
1 year	3.8	2.6	0.09
5 years	20.6	3.4	0.28

The accompanying figures relate dividend yield to subsequent long-horizon returns; predictability is stronger at a five-year horizon than one year in the reported regression. A scatter of log dividend yield against five-year return has a positive relation.

5.5 In-sample vs out-of-sample R^2

$$\text{In-sample } R^2 = 1 - \frac{\text{SSR}}{\text{SST}}$$

$$\text{SSR} = \sum (y_i - \hat{y}_i)^2; \text{SST} = \sum (y_i - \bar{y})^2; 0 \le R^2 \le 1$$

$$\text{OOS } R^2 = 1 - \frac{\sum_t (R^e_{m,t} - \hat{R}^e_{m,t})^2}{\sum_t (R^e_{m,t} - \bar{R}^e_{m,t})^2}$$

- $\hat{R}^e_{m,t}$ comes from a predictive regression estimated only through t−1.
- $\bar{R}^e_{m,t}$ is the historical average market excess return estimated only through t−1.
- OOS $R^2>0$: predictive model has lower average squared forecast error than historical mean; OOS $R^2<0$: it is worse.

5.6 Factor investing and characteristic forecasts

Tangency portfolio

$$w^* = \frac{\Sigma^{-1}\mu^e}{1^\top \Sigma^{-1}\mu^e}$$

Factor investing selects securities using characteristics associated with higher expected returns and/or lower risk. A one-characteristic prediction model is:

$$\mathbb{E}_t[R^e_{i,t+1}] = \delta_0 + \delta_1 z_{i,t}$$

Z-scores

$$z_i = \frac{x_i - \bar{x}}{\sigma_x}$$

- Standardization puts characteristics measured in different units onto a common scale; it also creates a natural rank within the cross-section.

Estimate

$$R_{i,t} - R_{f,t} = \delta_0 + \delta_1 z_{i,t-1} + \varepsilon_{i,t}$$

Forecast

$$\mathbb{E}_t[R^e_{i,t+1}] = \hat{\delta}_0 + \hat{\delta}_1 z_{i,t}$$

With multiple characteristics, include multiple z's and their matching coefficients.

notation conventions: The analysis's two-characteristic forecast prints δ_1 on both z_1 and z_2. The model logic requires the coefficient paired with each characteristic (δ_1 for z_1, δ_2 for z_2, etc.); remember the illustration typo rather than reproducing it in a calculation.

Building a factor portfolio

1. Choose and calculate firm characteristics.
2. Standardize (often z-score) and forecast/rank expected returns.
3. Sort securities into quantiles/deciles.
4. Form a long-short portfolio: long high-score / high-forecast group, short low-score group; weight legs equally or by market value depending on construction.

5.7 Cross-sectional styles and return signals

Style	Common variables / definition
Value	Market-to-book, price/earnings, price/dividend; "cheap" relative to fundamentals.
Size	Market equity / market capitalization.
Momentum	Prior return / recent relative performance.
Risk	Beta, total volatility, idiosyncratic volatility.
Investment	Asset growth; conservative vs aggressive investment.
Issuance	Net equity issuance.
Profitability / quality	Profitability, stability, growth, management quality.
Leverage	Balance-sheet leverage.
Liquidity	Trading activity / ease of trading.

Value vs growth sort

$$MB = \frac{\text{market equity}}{\text{book equity}}$$

- High MB = growth; low MB = value.
- A portfolio-sorting example uses firm characteristics measured at a formation date (e.g., December) and compares subsequent returns across sorted groups.

Historical factor-return patterns

Pattern	Observation message / associated reference
Value vs growth	Value has earned higher average long-run returns in the classic evidence; Fama & French (1992).
Small vs big	Small firms historically earned a size premium over big firms in long samples.
Profitability	High/robust profitability associated with higher average returns; Novy-Marx (2013).

Investment	Conservative asset growth associated with higher average returns than aggressive investment; Cooper, Gulen & Schill (2008).
Momentum	Recent winners outperform recent losers on average; Jegadeesh & Titman (1993); Asness, Moskowitz & Pedersen (2013).
Low variance	Low-volatility stocks can outperform high-volatility stocks on a risk-adjusted basis; Ang et al. (2006).

Recent ETF charts in the underlying analysis also show that style performance is regime-dependent; long-run premia do not imply every short sample favors the same style.

PART 6

Measuring Joint Risk: Covariance, Diversification, and Risk Models

Risk in a portfolio is joint rather than merely individual. Even if each security has nonzero variance, combining imperfectly correlated payoffs can lower portfolio variance. For a vector of asset returns R,

$$\Sigma_{ij} = \mathrm{Cov}(R_i, R_j), \qquad \mathrm{Var}(w^\top R) = w^\top \Sigma w.$$

Idiosyncratic fluctuations can diversify away, while a common component survives as the portfolio grows. A factor *risk* model replaces thousands of pairwise correlations with exposures to shared shocks and residual variances. Such a model is an estimator of Σ, not by itself a causal explanation for the expected premium on each factor.

Three distinct meanings of factor. An *alpha characteristic* (such as a valuation ratio) helps forecast μ; a *risk factor* (such as a common market shock) helps explain Σ; an *asset-pricing factor* is a systematically priced exposure supporting an equilibrium restriction on μ. One empirical quantity may enter more than one model, but its economic role must be stated explicitly.

6.1 Two risky assets: expected return, variance, covariance, correlation

$$R_p = wR_1 + (1-w)R_2$$

$$\mathbb{E}[R_p] = w\mu_1 + (1-w)\mu_2$$

$$V(R_p) = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\,\mathrm{Cov}(R_1, R_2)$$

Population covariance

$$\mathrm{Cov}(R_1, R_2) = \mathbb{E}[(R_1 - \mathbb{E}[R_1])(R_2 - \mathbb{E}[R_2])]$$

Sample covariance

$$\widehat{\mathrm{Cov}} = \frac{1}{T-1}\sum_t (R_{1,t} - \bar{R}_1)(R_{2,t} - \bar{R}_2)$$

Correlation

$$\rho_{12} = \frac{\mathrm{Cov}(R_1, R_2)}{\sigma_1\sigma_2}, \qquad -1 \le \rho \le 1$$

Correlation cases

ρ	**Portfolio risk / implication**
$+1$	$\sigma_p = w\sigma_1 + (1-w)\sigma_2$ for nonnegative weights. Risk is a straight-line combination; no covariance diversification benefit.
-1	$\sigma_p = \lvert w\sigma_1 - (1-w)\sigma_2 \rvert$. A zero-volatility mix is possible.
0	$\sigma_p = \sqrt{w^2 V(R_1) + (1-w)^2 V(R_2)}$. Curvature reflects diversification.
Between -1 and $+1$	Lower correlation bends the feasible set farther left, delivering lower risk for a given expected return.

Worked interview question — $\rho = -1$: Set $w\sigma_1 = (1-w)\sigma_2$ to get zero risk. Therefore

$$w = \frac{\sigma_2}{\sigma_1 + \sigma_2} \quad \textbf{in asset 1}, \qquad 1 - w = \frac{\sigma_1}{\sigma_1 + \sigma_2} \quad \textbf{in asset 2}.$$

6.2 Diversification vs concentration

Bessembinder evidence: most individual U.S. common stocks since 1926 have lifetime cumulative returns below one-month Treasury bills. In lifetime dollar wealth creation, the best-performing ~4% of listed companies explain the net gain of the entire U.S. stock market. Positive skew and compounding make concentrated stock-picking especially difficult; poorly diversified active strategies often underperform market averages.

Carhart (1997) is shown on persistence in mutual-fund performance: past relative performance does not translate into simple, dependable long-run persistence after accounting for known effects.

Diversification across asset classes — ETFs listed

Ticker	**Exposure**	**Ticker**	**Exposure**
EEM	MSCI Emerging Markets	EFA	MSCI EAFE
EMB	J.P. Morgan USD EM Bonds	EPP	Pacific ex Japan
EZA	South Africa	EZU	Eurozone
FXI	China Large-Cap	GSG	S&P GSCI commodities
IAU	Gold	ICF	REITs
IEF	7–10Y U.S. Treasury	ILF	Latin America 40
JPXN	JPX-Nikkei 400	LQD	Investment-grade corporate bonds
SPY	S&P 500	TLT	20+Y U.S. Treasury

Why diversification works

For an equal-weighted portfolio of N assets, define average individual variance and average pairwise covariance:

$$\bar{\sigma}^2 = \frac{1}{N}\sum_i \sigma_i^2$$

$$\overline{\mathrm{Cov}} = \frac{1}{N(N-1)}\sum_{i \neq j} \mathrm{Cov}(R_i, R_j)$$

$$V(R_p) = \frac{1}{N}\bar{\sigma}^2 + \frac{N-1}{N}\overline{\text{Cov}}$$

KEY: As N→∞, the average idiosyncratic-variance term goes to zero, but common covariance risk remains. Diversification eliminates firm-specific risk, not systematic risk.

Risk component	Meaning	Diversifiable?
Systematic risk	Market/economy-wide common risk.	No.
Idiosyncratic risk	Firm/security-specific risk.	Yes, increasingly as N grows.

6.3 CAPM as a risk model

$$V(R_i^e) = \beta_i^2 V(R_m^e) + V(\varepsilon_i)$$

- First term = systematic variance; second = firm-specific variance.

If residuals are mutually uncorrelated

$$\text{Cov}(R_i^e, R_j^e) = \beta_i \beta_j V(R_m^e)$$

KEY: A factor model can reduce the number of covariance inputs by explaining comovement through common factors instead of estimating every pairwise covariance separately.

6.4 Characteristic, risk-factor, and pricing-factor distinctions

Model type	Purpose	Generic form
Z-score / characteristic model	Forecast future expected returns from firm characteristics.	E[R_i^e]=δ₀+δ₁z_{1i}+δ₂z_{2i}+...
Factor model	Explain expected/realized returns through exposures to common factor returns.	E[R_i^e]=α_i+β_i^1F₁+β_i^2F₂+...

- Factors are often long-short portfolios that capture common risk premia.
- Performance attribution decomposes a portfolio's return into contributions from market, size, value, momentum, etc., showing what actually drove performance and whether the portfolio matches its intended strategy.
- Risk-factor data/model providers listed: MSCI Barra, Axioma, Northfield.

MSCI Barra US Equity Model — style factors listed

Style factor	Description
Beta / Non-Linear Beta	Components of risk driven by market sensitivity.
Liquidity	Return patterns associated with relative trading activity.
Residual Volatility	High-volatility return patterns not captured by beta.
Momentum	Return differences based on recent relative performance.
Size / Non-Linear Size	Systematic risk differences between large- and small-cap assets.
Book-to-Price, Growth, Leverage, Dividend Yield, Earnings Yield	Style premia associated with the firm's business model.

Covariance compression. With factor-return vector F, factor-loading matrix B, residual vector ε and mutually uncorrelated residuals, a standard risk-model representation is

$$R^e = BF + \varepsilon, \qquad \Sigma = B\,\mathrm{Var}(F)B^{\mathsf{T}} + \mathrm{Var}(\varepsilon).$$

This decomposition estimates a relatively small factor covariance matrix plus a residual variance structure, avoiding direct estimation of every pairwise covariance. In contrast, a pricing-factor model is a statement about expected compensation, not just covariance fitting.

PART 7

Time-Varying Risk: Volatility Forecasting

Portfolio risk changes through time because shocks tend to cluster. The information set at time t therefore contains useful information about $\mathrm{Var}_t(R_{t+1})$. This does not imply that the sign of the next market move can be forecast.

$$h_{t+1} = \omega + \alpha\varepsilon_t^2 + \beta h_t \qquad \text{(GARCH(1,1))}.$$

The intercept anchors longer-run variation, α measures sensitivity to new squared shocks, and β measures persistence. Risk forecasting changes the denominator of the portfolio exposure rule, whereas expected-return forecasting changes its numerator. Hence a market can be difficult to time by direction but still allow predictable risk scaling.

7.1 Conditional volatility models

$$v_t^* = \frac{\mathbb{E}[R_m^e]}{\gamma V_t(R_{m,t+1})}$$

When expected excess return is held fixed but conditional variance changes, the optimal rule reduces equity exposure in high-uncertainty periods and increases exposure in low-volatility periods. The analysis cites Moreira & Muir (2017), Volatility-Managed Portfolios.

ARCH/GARCH motivation

- ARCH: Engle (1982); GARCH: Bollerslev (1986). Conditional volatility is time-varying and persistent, and large shocks raise future volatility.
- Models capture clusters of turbulence and calm; many extensions allow multiple assets, asymmetric responses, and richer lag structures.
- The analysis names the Heterogeneous Autoregressive (HAR) realized-volatility model of Corsi (2009) as a better model.

Exponentially weighted moving average (EWMA)

Let $h_t = \hat{\sigma}_t^2$ denote conditional variance and $\alpha \in (0,1)$ be the weight on the newest squared return innovation:

$$h_t = \alpha R_{t-1}^2 + (1-\alpha)h_{t-1}, \qquad \widehat{h}_{t+1} = \alpha R_t^2 + (1-\alpha)h_t.$$

These are the same recursion, shifted one period. A larger α gives more weight to new volatility observations. Illustrative newest-observation weights are $\alpha = 0.06$ for daily returns and $\alpha = 0.03$ for monthly returns. Another widespread convention defines the persistence parameter $\lambda = 1 - \alpha$ instead, so that $h_{t+1} = (1-\lambda)R_t^2 + \lambda h_t$. The numerical weights depend on which coefficient the notation designates.

```
returns.ewm(alpha=<alpha>).std()
```

GARCH(1,1)

$$R_t = \mu + u_t, \quad u_t = \sigma_t z_t, \quad z_t \sim N(0,1)$$
$$\sigma_t^2 = k + \alpha u_{t-1}^2 + \beta\sigma_{t-1}^2$$

- α captures shock impact; β captures variance persistence.

$$\hat{\sigma}_{t+1}^2 = k + \alpha u_t^2 + \beta \sigma_t^2$$

$$v_t^* = \frac{\mu - \bar{R}_f}{\gamma \hat{\sigma}_{t+1}^2}$$

Maximum likelihood for GARCH

$$L(\theta) = \sum_t \left[-\frac{1}{2}\ln(2\pi) - \frac{1}{2}\ln(\sigma_t^2) - \frac{(R_t - \mu)^2}{2\sigma_t^2} \right],$$

$$\theta = (\mu, k, \alpha, \beta)$$

1. Choose initial guesses for θ.
2. Recursively compute σ_t².
3. Maximize log-likelihood over θ.

```
pip install arch

from arch import arch_model

model = arch_model(returns, vol='Garch', p=1, q=1)

fitted_model = model.fit()

print(fitted_model.summary())
```

Empirical VTI GARCH(1,1) fit

Parameter	Estimate	Interpretation
μ	7.3738e−4	Mean daily return in the fitted constant-mean specification.
ω (k)	2.8776e−6	Variance intercept.
α[1]	0.1000	Shock loading.
β[1]	0.8800	High variance persistence.

6342 observations; normal innovations. α+β≈0.98 signals very persistent volatility. The accompanying conditional-volatility plot spikes in crisis episodes.

Heston stochastic volatility

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^S$$

$$dv_t = \kappa(\theta - v_t)dt + \sigma\sqrt{v_t} dW_t^v; \mathrm{Corr}(dW^S, dW^v) = \rho$$

- ν_t mean-reverts to long-run θ at rate κ. The model captures volatility clustering and leverage effects; estimation can use ML or Bayesian methods.
- The analysis characterizes GARCH as a discrete-time approximation to Heston-style stochastic volatility.

Heston–Nandi GARCH

$$R_t = \mu + u_t, \quad u_t = \sigma_t z_t, \quad z_t \sim N(0,1)$$
$$\sigma_t^2 = \omega + \beta\sigma_{t-1}^2 + \alpha(\varepsilon_{t-1} - \gamma\sigma_{t-1})^2$$

- γ captures leverage/asymmetry.

$$\hat{\sigma}_{t+1}^2 = \omega + \beta\sigma_t^2 + \alpha(\varepsilon_t - \gamma\sigma_t)^2$$
$$v_t^* = \frac{\mu - \bar{R}_f}{\gamma\hat{\sigma}_{t+1}^2}$$

PART 8

Portfolio Choice I: Choosing the Risky Composition

Given expected returns μ and joint risk Σ, first solve the *composition* problem among risky securities. A budget-feasible composition w satisfies $\mathbf{1}^\top w = 1$. Efficient portfolios have the minimum possible variance for their target expected return. Without a risk-free asset, this generates the risky-asset frontier.

$$\min_w \frac{1}{2} w^\top \Sigma w \quad \text{subject to} \quad \mathbf{1}^\top w = 1,\ \mu^\top w = \mu_p.$$

Introducing a risk-free asset makes the slope of every capital allocation line equal to its underlying risky portfolio's Sharpe ratio. Choosing the maximum slope identifies w_T, a *composition* of risky assets independent of individual risk aversion:

$$\boxed{w_T = \frac{\Sigma^{-1}(\mu - R_f \mathbf{1})}{\mathbf{1}^\top \Sigma^{-1}(\mu - R_f \mathbf{1})}}.$$

This normalization presumes the denominator is nonzero and short-sale/borrowing constraints do not bind. The Σ^{-1} operation penalizes exposures which add correlated risk relative to their forecast premium.

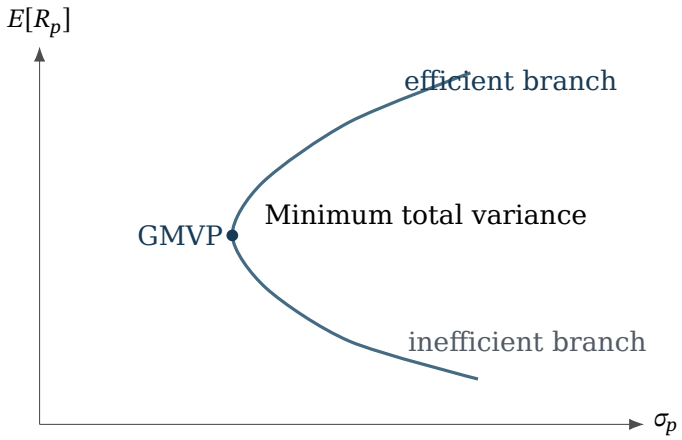


Figure 4: Risky-asset efficient frontier and the global minimum-variance portfolio (schematic).

8.1 Markowitz and asset allocation

The Nobel analysis highlights the 1990 economics prize to Harry Markowitz (portfolio choice), William Sharpe (asset pricing/CAPM), and Merton Miller (corporate finance). The portfolio-choice problem asks how to allocate wealth across risky asset classes, first with two assets, then N assets, then N risky assets plus a risk-free asset.

8.2 Portfolio frontier with risky assets

- The feasible frontier plots expected return against standard deviation for all weight combinations.

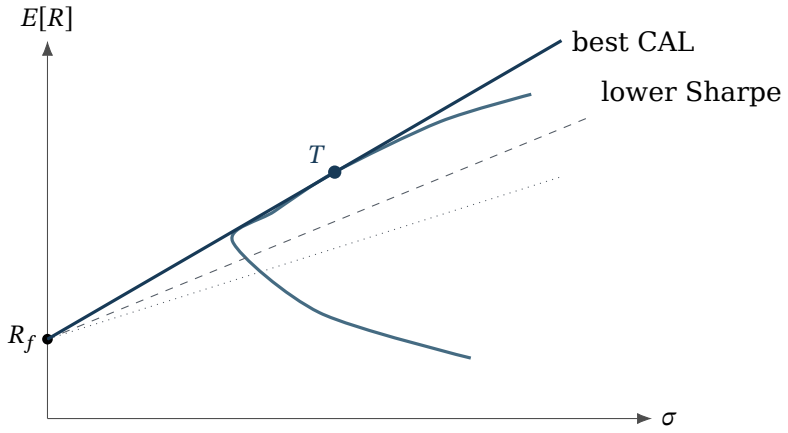


Figure 5: Among risk-free–risky combinations, the optimal CAL has maximum Sharpe slope and touches the risky frontier at T (schematic).

- Global minimum-variance portfolio (GMVP/MVP) is the left-most feasible portfolio.
- Portfolios above the MVP on the upper branch are mean–variance efficient; those below it are inefficient because another portfolio offers higher return for the same risk.
- A no-short-selling constraint shrinks the feasible set and can shift the frontier inward relative to the unconstrained frontier.

8.3 N risky assets: matrix formulation and closed-form frontier

$$R_p = \sum_i w_i R_i; \mathbb{E}[R_p] = w^\top \mu; V(R_p) = w^\top \Sigma w$$

Efficient-frontier optimization for a target expected return R_p*:

$$\min_w \ \tfrac{1}{2} w^\top \Sigma w \quad \text{s.t.} \quad R_p^* = w^\top \mu, w^\top 1_N = 1$$

$$w^* = \lambda_1 \Sigma^{-1}\mu + \lambda_2 \Sigma^{-1} 1_N$$

$$A = \mu^\top \Sigma^{-1}\mu, B = \mu^\top \Sigma^{-1} 1_N, C = 1_N^\top \Sigma^{-1} 1_N, D = AC - B^2$$

$$\lambda_1 = \frac{CR_p^* - B}{D}; \qquad \lambda_2 = \frac{A - BR_p^*}{D}$$

$$\sigma_p^2 = \frac{A - 2BR_p^* + C(R_p^*)^2}{D}$$

Global minimum-variance portfolio

$$\min_w \ \tfrac{1}{2} w^\top \Sigma w \quad \text{s.t.} \quad 1_N^\top w = 1$$

$$w_{\text{MVP}}^* = \frac{\Sigma^{-1} 1_N}{C}, \quad \text{where} \quad C = 1_N^\top \Sigma^{-1} 1_N$$

Why the frontier is a quadratic curve. The Lagrange first-order condition is $\Sigma w = \lambda_1 \mu + \lambda_2 \mathbf{1}$. Solving $\mu^\top w = \mu_p$ and $\mathbf{1}^\top w = 1$ for λ_1, λ_2 gives the coefficients A, B, C, D above. Substituting these multipliers back into $w^\top \Sigma w$ yields $\sigma_p^2 = (A - 2B\mu_p + C\mu_p^2)/D$. This equation describes the entire unconstrained minimum-variance boundary, not just one portfolio.

8.4 Add the risk-free asset: CAL and tangency portfolio

Let v be invested in a risky portfolio R_p and 1−v in the risk-free asset:

$$R_{\text{CAL}} = \upsilon R_p + (1-\upsilon)R_f = R_f + \upsilon(R_p - R_f)$$
$$\mathbb{E}[R_{\text{CAL}}] = R_f + \upsilon(\mathbb{E}[R_p] - R_f); \sigma(R_{\text{CAL}}) = \upsilon\sigma(R_p)(\upsilon \geq 0)$$

CAL

$$\mathbb{E}[R_{\text{CAL}}] = R_f + \frac{\mathbb{E}[R_p] - R_f}{\sigma(R_p)}\sigma(R_{\text{CAL}})$$

Sharpe ratio

$$\text{SR} = \frac{\mathbb{E}(R_p) - R_f}{\sigma(R_p)}$$

• The optimal CAL uses the risky tangency portfolio that maximizes Sharpe ratio.

• Two-fund separation: every investor holds the same tangency portfolio of risky assets plus the risk-free asset; only v differs by risk tolerance.

Sharpe-ratio maximizing portfolio

$$\max_{w} \frac{w^{\top}\mu - R_f}{\sqrt{w^{\top}\Sigma w}}$$

Normalized solution

$$w^*_{\text{SR}} = \frac{\Sigma^{-1}(\mu - 1_N R_f)}{1_N^{\top}\Sigma^{-1}(\mu - 1_N R_f)}$$

Equivalent minimum-variance formulation for a target return with risk-free borrowing/lending:

$$\min_{w} \ \tfrac{1}{2} w^{\top}\Sigma w \quad \text{s.t.} \quad R_p^* = w^{\top}\mu + (1 - w^{\top}1_N)R_f$$

Portfolio optimization tools and applications

• PyPortfolioOpt implements efficient frontiers, Black–Litterman allocation, covariance shrinkage, Hierarchical Risk Parity, and exponentially weighted covariance estimates.

• Wealthfront provides an applied example of automated portfolio allocation.

• Portfolio-management software translates target weights into investable orders and rebalancing decisions.

PART 9

Portfolio Choice II: Choosing the Risk Scale

The portfolio w_T identifies the most attractive risky *fund*; it does not specify how much wealth an investor allocates to that fund. Let y be the total risky allocation and $x = yw_T$ be the vector of final risky weights. The risk-free weight is $1 - y$.

$$R_C = R_f + yR_T^e, \quad E[R_C] = R_f + y\mu_T^e, \quad \text{Var}(R_C) = y^2\sigma_T^2.$$

Maximizing $U(y) = R_f + y\mu_T^e - (\gamma/2)y^2\sigma_T^2$ yields

$$\frac{\partial U}{\partial y} = \mu_T^e - \gamma y\sigma_T^2 = 0, \qquad \boxed{y^* = \frac{\mu_T^e}{\gamma\sigma_T^2}}.$$

Combining composition and scale eliminates the intermediate normalization:

$$\boxed{x^* = y^* w_T = \frac{1}{\gamma}\Sigma^{-1}(\mu - R_f\mathbf{1})}.$$

Two different *tangencies* are involved: the best CAL touches the *efficient frontier* to determine w_T; the investor's highest indifference curve touches that *CAL* to determine y^*. Confusing these points loses the separation between opportunity and preferences.

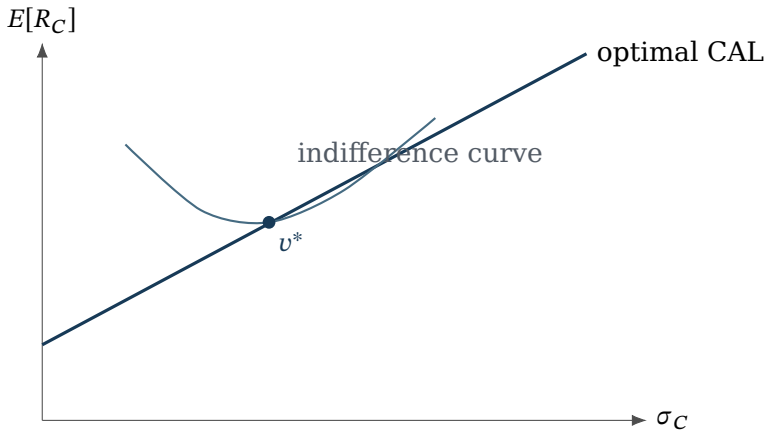


Figure 6: Risk-aversion determines the preferred position on the best CAL (illustrative geometry).

9.1 Mean–variance preferences and indifference curves

$$U(R) = \mathbb{E}[R] - \frac{\gamma}{2}V(R)$$

- γ is the risk-aversion coefficient. γ=0 → risk-neutral in this mean–variance objective.
- Utility rewards expected return and penalizes variance. What is missing? Higher moments / non-mean–variance aspects of the return distribution.

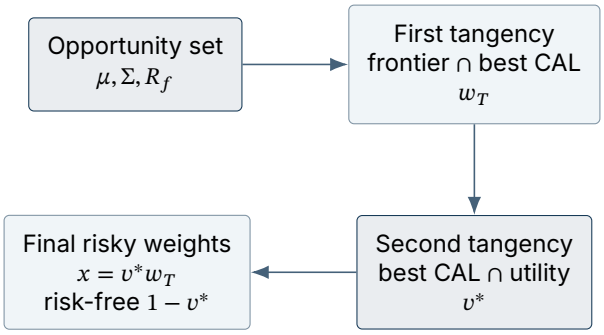


Figure 7: The risky composition w_T and risky scale v answer different questions.

Derived shape of indifference curves: For constant U: E[R]=U+(γ/2)σ². In (σ,E[R]) space, each indifference curve is upward-sloping and convex; higher curves mean higher utility; larger γ makes the investor require more expected return for a given increase in risk.

9.2 Risky-fund and risk-free composition

$$R_p = vR_m + (1 - v)R_f = R_f + vR_m^e$$
$$\mathbb{E}[R_p] = R_f + vE[R_m^e]$$
$$V(R_p) = v^2 V(R_m); \sigma(R_p) = |v|\sigma(R_m)$$

Capital Allocation Line (for v≥0)

$$\mathbb{E}[R_p] = R_f + \frac{\mathbb{E}(R_m^e)}{\sigma(R_m)}\sigma(R_p)$$

- CAL slope = Sharpe ratio = expected excess return per unit of total risk.
- Optimal asset allocation is the point where the highest attainable indifference curve is tangent to the CAL. More risk-averse → more risk-free asset; more risk-tolerant → more risky asset.

9.3 Optimal risk exposure, leverage, and constraints

Static unconstrained optimum

$$v^* = \frac{\mathbb{E}[R_m^e]}{\gamma V(R_m)}$$

Under the static benchmark, if stock returns are IID, the investor treats the expected excess return and variance as constant, so the optimal risky weight is static. In practice, managers update portfolios as new information changes conditional expected returns or risk.

Dynamic optimum

$$v_t^* = \frac{\mathbb{E}_t[R_{m,t+1}^e]}{\gamma V_t(R_{m,t+1})}$$

- Risk aversion γ itself could vary over time.

Many real portfolio managers face 0≤v≤1: no borrowing/leverage and no short position in the risky asset. The weights have the following interpretations:

Weight	Meaning
v=0	100% risk-free.
v=1	100% risky.
v>1	Leveraged: borrow to invest more than wealth in risky asset.
v<0	Short risky asset; invest more than 100% in risk-free asset.

Institutional reasons: regulation, risk-management policy, fiduciary responsibilities, margin requirements, and borrowing costs.

Worked leverage examples: With W_0=$100,000: v=1.5 means invest $150,000 in risky asset and borrow $50,000 at R_f. v=−0.5 means short $50,000 of risky asset and invest $150,000 in risk-free asset.

PART 10

Dynamic Asset Allocation

Static portfolio theory becomes a conditional policy when expected returns and conditional variances change. For a fixed risky fund,

$$\boxed{v_t^* = \frac{E_t[R^e_{T,t+1}]}{\gamma \operatorname{Var}_t(R_{T,t+1})}}.$$

The numerator is *return timing*: adjust risk when the forecasted reward changes. The denominator is *volatility timing*: reduce risk when conditional variance increases, holding risk aversion and expected premium fixed. Both are applications of the same utility optimization problem, not unrelated trading tricks.

Timing rules must use contemporaneously observable signals and investable prices. The ex post reward to a perfect timer is an unattainable informational upper bound; backtests that identify turning points only after the event are not evidence of predictability. Regression-based convexity tests such as Treynor–Mazuy measure timing-like nonlinear exposure but need careful interpretation in the presence of options or nonlinear securities.

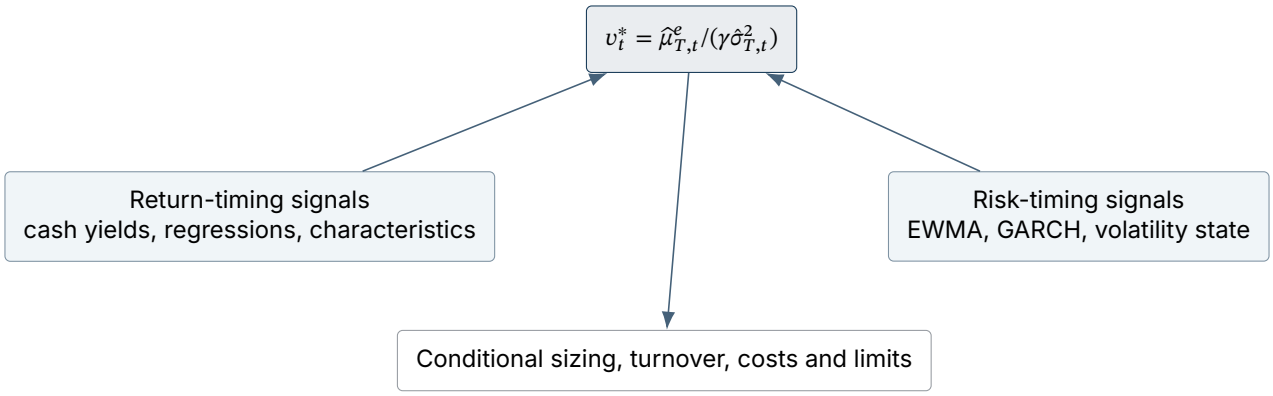


Figure 8: Dynamic timing changes the numerator and/or denominator of the same exposure rule.

10.1 Random walk and market timing

Many investors search for predictable price patterns. The economic intuition is that near-random price changes are a consequence of intelligent investors competing to exploit information; prices react rapidly to new information.

Market timing shifts capital between market and safe asset based on expected market outperformance. Signals/examples listed: "Sell in May and go away," moving averages, Twitter sentiment, Google Trends, macro signals.

Seasonality visual: The "Unkindest Month" chart compares average monthly DJIA percentage changes over 100, 50, and 20 years. September is often the weakest month; the chart is an example of a historical seasonal pattern, not proof of an exploitable free lunch.

Cost of missing the best days (Tiingo daily data, Aug. 2002–present)

Scenario	QQQ	SPY	LQD	IEF
All data	.1591	.1122	.0460	.0352
Missing best 5 days	.1348	.0877	.0320	.0294
Missing best 10	.1184	.0727	.0249	.0254
Missing best 20	.0916	.0491	.0163	.0189
Missing best 30	.0690	.0298	.0093	.0130

KEY: A market-timing strategy can lose a large fraction of long-run performance by being out of the market for only a small number of the best days.

10.2 Predictive market-timing implementation

Estimate using only information available through t

$$R^e_{m,\tau} = a_t + b_t F_{\tau-1} + \varepsilon_\tau, \tau = 1, \dots, t$$

Forecast

$$\hat{R}^e_{m,t+1} = \hat{a}_t + \hat{b}_t F_t$$

Weight

$$v^*_t = \frac{\hat{R}^e_{m,t+1}}{\gamma V(R_m)}$$

Realized portfolio

$$R_{p,t+1} = v^*_t R_{m,t+1} + (1 - v^*_t) R_{f,t+1}$$

- At t+1, repeat with the enlarged data set. This is a rolling/expanding real-time experiment.

Static IID investor

$$v^* = \frac{\mathbb{E}[R^e_m]}{\gamma V(R_m)}$$

Perfect market timer benchmark

The analysis's static investor uses γ≈4 so v*=0.6. Under a no-leverage/no-short market-timing interpretation, a perfect timer would hold whichever of market or safe asset is higher ex post each period: R_PMT,t=max(R_m,t,R_f,t). This is perfect foresight—not a feasible strategy.

	R_m	R_f	Static R_p	Perfect timer
Mean	.0990	.0459	.0778	.2590
Std. dev.	.1404	.0444	.0849	.0895
Sharpe	.3783	0	.3754	2.3807

Monthly min	−.2619	−.0826	−.1551	−.0347
Monthly max	.5131	.1090	.3100	.5131

Robert Shiller data, 1871–present. Observations: R_m 1853; R_f 1854; static and PMT 1853.

10.3 Detecting market timing with Treynor–Mazuy

$$R_{p,t} - R_{f,t} = \alpha + \beta_1(R_{m,t} - R_{f,t}) + \beta_2(R_{m,t} - R_{f,t})^2 + \varepsilon_t$$

- If a manager increases beta before up markets and reduces beta before down markets, the payoff relation becomes convex.
- $\beta_2 > 0$ is evidence consistent with successful market timing.

PITFALL — look-ahead bias / data leakage: Backtests must use only information available to investors on that date. Do not run predictive regressions once on the full data set and then pretend those coefficients were known in the past.

PART 11

Equilibrium Asset Pricing: Portfolio Choice to CAPM

The CAPM is an equilibrium result obtained by combining optimal *individual* choices with *market clearing*. With identical beliefs about μ, Σ, and R_f, every investor chooses the same tangency composition w_T, though investor h holds a different dollar amount $v_h W_h$.

$$\text{Investor } h: \quad q_{h,i} = v_h W_h w_{T,i}, \qquad \sum_h q_{h,i} = w_{T,i} \sum_h v_h W_h.$$

Equating this aggregate demand to the outstanding market value of each security implies $w_T = w_m$, where $w_{m,i} = ME_i / \sum_j ME_j$. Therefore the market portfolio is the equilibrium tangency fund.

Unnormalized mean–variance demand is proportional to $\Sigma^{-1}\mu^e$. Market clearing implies $\mu^e = k\Sigma w_m$ for a positive aggregate risk-price coefficient k. Multiplying by $w_m^\top$ gives $\mu_m^e = k\sigma_m^2$, and therefore

$$\boxed{E[R_i] - R_f = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)} \big(E[R_m] - R_f\big)}.$$

The *security market line* prices beta, not volatility. The *capital market line* relates expected return to *total volatility* only for efficient combinations of market and risk-free assets. An arbitrary risky portfolio lies on neither line merely because it has a certain standard deviation.

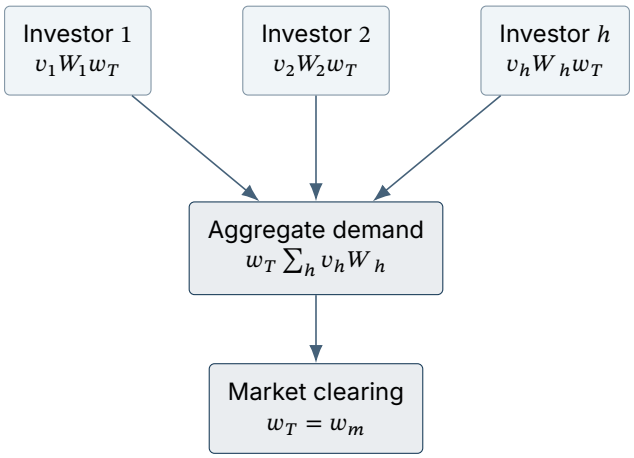


Figure 9: CAPM market clearing converts the common individually optimal risky fund into the value-weighted market portfolio.

11.1 CAPM assumptions and equilibrium market portfolio

CAPM links expected return to systematic risk measured by beta. The model assumes investors:

1. are rational and use portfolio theory
2. have the same information
3. can buy and short-sell any asset

4. are price takers / do not affect security prices
5. have the same investment horizon
6. pay no taxes or transaction costs
7. differ only in risk aversion γ

In equilibrium, all investors hold the same tangency portfolio of risky assets; aggregate supply then forces that tangency portfolio to be the market portfolio.

Market value weight

$$w_i = \frac{\text{market value of firm } i}{\text{market value of all firms}}$$

• This is value weighting. Alternatives mentioned: equal weighting and price weighting.

• The Capital Market Line (CML) connects R_f to the market portfolio. More risk-averse investors hold more risk-free asset; more risk-tolerant investors hold more market portfolio, possibly levered in the unconstrained model.

11.2 Deriving the capital asset pricing model

With N risky assets, expected excess-return vector μ^e, covariance Σ, and mean–variance risk aversion γ:

Optimal risky weights

$$w = \frac{1}{\gamma}\Sigma^{-1}\mu^e$$

In equilibrium, the investor must hold the market portfolio w_m:

$$w_m = \frac{1}{\gamma}\Sigma^{-1}\mu^e \quad \Rightarrow \quad \mu^e = \gamma\Sigma w_m$$

For asset i

$$\mathbb{E}[R_i] - R_f = \gamma\,\mathrm{Cov}(R_i, R_m)$$

For the market

$$\gamma = \frac{\mathbb{E}(R_m) - R_f}{V(R_m)}$$

CAPM

$$\mathbb{E}[R_i] - R_f = \beta_i[\mathbb{E}(R_m) - R_f]$$

$$\beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{V(R_m)}$$

Security Market Line (SML)

$$\mathbb{E}[R_i] = R_f + \beta_i(\mathbb{E}[R_m] - R_f)$$

The SML plots expected return against beta, not total volatility. Numerical example: with R_f=3% and market expected return 9%, E[R_i]=3%+β_i×6%.

CAPM implications

- All investors hold the market portfolio of risky securities.
- Firm-specific/idiosyncratic risk does not increase expected return because it can be diversified away.
- Only systematic risk affects the equilibrium discount rate and valuation; beta measures that systematic exposure.

Reality check: CAPM relies on stylized assumptions: actual markets have heterogeneous information/horizons, constraints, taxes, transaction costs, illiquidity, and other frictions. "All models are wrong, but some are useful."

PART 12

Alpha, Beta, and Testing Asset-Pricing Models

Once a pricing model is specified, its predictions can be confronted with realized returns. A CAPM time-series regression takes the form

$$R^e_{i,t} = \alpha_i + \beta_i R^e_{m,t} + \varepsilon_{i,t}.$$

The fitted beta measures historical comovement with the market; the intercept alpha is the abnormal return *relative to this benchmark model*. A statistically significant alpha could indicate a true pricing anomaly, frictions, a missing risk factor, or statistical/sample effects. Realized market outperformance by itself is not synonymous with positive alpha.

12.1 Testing CAPM: alpha and beta

Expected excess return with alpha

$$\mathbb{E}[R^e_i] = \alpha_i + \beta_i E[R^e_m]$$

Time-series regression

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i (R_{m,t} - R_{f,t}) + \varepsilon_{i,t}$$

• α_i = abnormal average return unexplained by CAPM. β_i = market sensitivity. ε_i,t = idiosyncratic deviation.

• A standard test rejects CAPM for an asset/portfolio if α is statistically different from zero (e.g., p<5%).

Amazon example: sample moments

The cumulative-return and excess-return charts preceding the regression show AMZN moving strongly with the market but with much larger swings; the AMZN-vs-market excess-return scatter has a clear positive slope, consistent with a beta above one.

	AMZN	R_m	R_f	R_m−R_f
Observations	60	60	60	60
Annual mean	.1416	.1633	.0271	.1362
Annual std. dev.	.3155	.1682	.0067	.1679
Monthly min	−.2375	−.0941	0	−.0941
Monthly max	.2706	.1246	.0048	.1245

Amazon CAPM regression

Term	Coef.	Std. err.	t	p-value	Interpretation
α / const	−0.0020	0.009	−0.222	0.825	−20 bp/month; not statistically significant.
Market β	1.4529	0.190	7.635	<0.001	A +1% market excess return is associated with about +1.45% AMZN excess return.

R^2=0.501: about half of the time-series variation in AMZN excess returns is explained by market excess return in this sample; the rest is residual/idiosyncratic variation.

Python pattern

```
# obtain asset/market/risk-free data (Tiingo + Ken French in observations)

import statsmodels.api as sm

y = asset_return - rf

X = sm.add_constant(market_return - rf)

reg = sm.OLS(y, X).fit()

print(reg.summary())
```

12.2 Alpha as mispricing and a Gordon-growth worked interpretation

• α>0: asset earned more than CAPM predicts → interpreted as undervalued / buy signal in the illustration's mispricing framing.

• α<0: asset earned less than CAPM predicts → interpreted as overvalued / sell signal.

• Fundamental analysts and quantitative regression-based analysts are two routes to estimating mispricing.

Observation example inputs: D_1=\$50/share, dividend growth g=8%, α=1%, β=1.5, R_f=3%, market excess premium=6%. One internally consistent way to connect these numbers is:

$$\text{CAPM required return} = 3\% + 1.5 \times 6\% = 12\%$$

$$\text{If } \alpha = +1\%, \quad \text{expected actual return} = 13\%$$

Price delivering 13% in Gordon model

$$P_{\text{market}} = \frac{50}{0.13 - 0.08} = \$1{,}000$$

CAPM fair price

$$P_{\text{CAPM}} = \frac{50}{0.12 - 0.08} = \$1{,}250$$

$$\text{Implied undervaluation / gain to CAPM fair value} = \$250/\text{share}$$

This is a worked interpretation of the illustration inputs; the illustration itself poses the "what trade / gain?" question rather than printing a full solution path.

12.3 Why CAPM can fail

Empirical “alpha” relative to CAPM can mean several things: genuine market inefficiency/behavioral bias; trading or liquidity frictions; or simply a missing risk factor. A significant alpha is therefore a joint test of the pricing model and market efficiency.

The empirical evidence summarizes value and size patterns as either compensation for omitted systematic risk (beta to another factor) or mispricing (alpha).

PART 13

Multifactor Asset Pricing

The same state-pricing argument permits multiple risk sources. Suppose, for transparent exposition, $M = a + \sum_k b_k F_k$ and the factors are normalized or orthogonalized. Substitution into the expected-excess-return identity gives

$$E[R_i^e] = -(1 + R_f) \sum_k b_k \operatorname{Cov}(F_k, R_i^e).$$

Define a quantity of exposure and a price per unit of factor exposure as

$$\beta_i^k = \frac{\operatorname{Cov}(R_i^e, F_k)}{\operatorname{Var}(F_k)}, \qquad \lambda_k = -(1 + R_f) b_k \operatorname{Var}(F_k).$$

Then

$$\boxed{E[R_i^e] = \sum_k \beta_i^k \lambda_k}.$$

A large β_i^k is not intrinsically good or bad: its required compensation depends on λ_k, which can be positive, zero, or negative. The CAPM is the special case where the only priced factor is the market excess return.

13.1 SDF view of factor models

$$1 = \mathbb{E}[M(1 + R_i)]$$

$$\mathbb{E}[1 + R_i] = (1 + R_f)(1 - \operatorname{Cov}(M, R_i))$$

$$\mathbb{E}[R_i^e] = -(1 + R_f) \operatorname{Cov}(M, R_i^e)$$

This can be represented as factor betas times factor prices of risk:

$$\mathbb{E}[R_i^e] = \beta_i \lambda$$

CAPM is the one-factor special case in which M is affine in market excess return:

$$M = a + b R_m^e$$

Multifactor SDF

$$M = a + \sum_{k=1}^{K} b_k F_k$$

Expected excess return

$$\mathbb{E}[R_i^e] = \sum_k \beta_i^k \lambda_k$$

For this exposition, assume factor innovations can be normalized/orthogonalized for exposition. Factors may be macro variables or traded factor portfolios.

Sign of a factor premium: Using the SDF equation: a factor/security that loses in bad times (high M) has negative Cov(M,R) and therefore a positive expected premium; an asset that hedges bad times can carry a negative premium; an unpriced factor can have zero premium.

13.2 Chen–Roll–Ross macro factors

Factor	Meaning
IP	Industrial-production growth / innovation.
EI	Expected inflation.
UI	Unexpected inflation.
CG	Corporate-bond vs long-government-bond excess return / credit-related spread factor.
GB	Long- vs short-government-bond excess return / term factor.

These variables illustrate the Arbitrage Pricing Theory idea that multiple systematic macro innovations can be priced.

PART 14

Fama–French and Modern Factor Investing

A characteristic is a number attached to a company or security, whereas a factor return is the realized payoff of a portfolio constructed from securities. These are related but not interchangeable. For example,

$$\boxed{\begin{array}{c}\text{high book-to-market characteristic} \to \text{cross-sectional sort} \\ \to \text{HML long–short return} \to \beta_{\text{HML}} \text{ exposure}\end{array}}.$$

Market, size, and value factor returns can be used jointly to describe realized returns and test whether an alpha remains after conditioning on multiple systematic styles. Additional profitability, investment, and momentum factors offer different dimensions of interpretation and are not replacements for careful forecast validation.

14.1 Fama–French three-factor model

$$\mathbb{E}[R_i^e] = \beta_{i,m}\mathbb{E}[R_m^e] + \beta_{i,\text{SMB}}\mathbb{E}[R_{\text{SMB}}] + \beta_{i,\text{HML}}\mathbb{E}[R_{\text{HML}}]$$

Factor	Construction / interpretation
Mkt−RF	Market excess return.
SMB	Small Minus Big: return of small-cap minus big-cap portfolio.
HML	High Minus Low book-to-market: value minus growth portfolio.

Regression form and interpretation

$$R_{i,t}^e = \alpha_i + \beta_m \text{MKT}_t + \beta_{\text{SMB}}\text{SMB}_t + \beta_{\text{HML}}\text{HML}_t + \varepsilon_{i,t}$$

- α: residual abnormal return / model mispricing.
- Market beta: sensitivity to broad market.
- SMB beta: positive → small-cap tilt; negative → large-cap tilt.
- HML beta: positive → value tilt; negative → growth tilt. Observation connects value exposure to systematic financial-distress risk.
- Reject a factor model for a test asset/portfolio when alpha is statistically different from zero under the chosen significance threshold.

Amazon Fama–French 3-factor regression

Term	Coef.	p-value	Interpretation
α	0.0055	0.491	Not significant.
Mkt−RF	1.2203	<0.001	Strong market exposure.
SMB	0.3702	0.203	Positive but not statistically significant in this sample.

HML	−0.9062	<0.001	Strong growth / negative value exposure.

R^2=0.646, up from the CAPM regression's 0.501 for the same 60-observation Amazon sample.

Characteristic to factor return. The book-to-market ratio is an observable characteristic. Ranking stocks by this characteristic and buying the high group while shorting the low group creates a time series of HML portfolio returns. Regressing a security's excess returns on this HML series measures an exposure, β_{HML}. Observing a high book-to-market ratio for one security does not itself determine its HML beta.

14.2 Fama–French extensions / additional factors

Factor	Definition
RMW	Robust Minus Weak profitability.
CMA	Conservative Minus Aggressive asset growth / investment.
MOM	High minus low prior returns (momentum).

PART 15

Long–Short Investing, Market Neutrality, and Economic Stories

A long–short investment can seek returns from relative valuation while reducing exposure to broad market moves. Dollar neutrality (total long dollars equal total short dollars) does not guarantee *beta* neutrality (aggregate sensitivity to market excess return equal to zero). If w_L and w_S denote signed weights, beta neutrality requires $\sum_i w_i \beta_i = 0$ rather than only $\sum_i w_i = 0$.

The economic interpretation of factor payoffs is generally not unique. A value or momentum premium may reflect hard-to-diversify risk, bounded rationality, slow information diffusion, and constraints on investors who would otherwise arbitrage away return differences. The quantitative strategy and the model explaining why it works should be distinguished.

15.1 Investment-style economic stories

Style	Mechanism emphasized on observations
Value	Buy securities cheap relative to fundamentals. Premium can be compensation for risk and/or correction of investor overreaction.
Trend following / momentum	Long rising assets, short falling assets. Possible drivers: underreaction, herding, and capital flows.
Liquidity provision	Provide liquidity to investors demanding immediacy; illiquid assets can offer a premium for bearing liquidity risk.
Low risk / betting against beta	Favor safer/lower-beta securities and use leverage if allowed. Leverage constraints can make high-beta assets overpriced and low-beta assets earn better risk-adjusted returns.
Quality	Favor profitable, stable, growing, well-managed companies; slow capital adjustment can sustain the premium.

15.2 Long-short and market-neutral investing

• Long = buy an asset expected to rise / be undervalued. Short = sell borrowed asset expected to fall / be overvalued.

• Market-neutral portfolio balances long and short positions—often equal gross dollar amounts or otherwise beta-neutral—to make net market beta approximately zero.

• Objective is alpha from relative security selection while hedging systematic market direction; in principle a correctly selected long-short spread can make money in bull or bear markets.

PART 16

Estimation Error and Robust Portfolio Construction

Optimal weights can be fragile even when the mathematical optimization problem is perfectly specified. Estimating N means and an $N \times N$ covariance matrix from limited histories creates many opportunities for sampling error; optimization then magnifies favorable noise in $\hat{\mu}$ and unfavorable noise in $\hat{\Sigma}$.

$$\hat{w}_T \propto \hat{\Sigma}^{-1}\hat{\mu}^e.$$

The answer to estimation risk may be a lower-variance estimator, stronger economically justified constraints, or a simpler rule such as $1/N$. The model selection problem therefore balances theoretical optimality against sensitivity, turnover, and out-of-sample stability. Simplicity is a defensible statistical choice rather than a failure to optimize.

16.1 Asset-class data, index funds, and implementation

Statistic	SPY	SHY	IEF	LQD
Observations	289	289	289	289
Annual mean	.1123	.0196	.0336	.0438
Annual risk	.1469	.0148	.0663	.0785
Annual downside risk	.0248	.0019	.0102	.0130
Ex. skewness	−.5609	.3125	.0822	.1193
Kurtosis	1.2482	2.0401	.7403	6.2534

Tiingo, August 2002–present (monthly sample in the illustration).

An index fund is a mutual fund or ETF designed to track an index such as the S&P 500. The S&P 500 measures performance of 500 large U.S. public companies by market capitalization.

Mutual fund	ETF
Trades at end-of-day NAV.	Trades continuously intraday at market prices; price may deviate from NAV.
Share sales can force realization/distribution of capital gains to shareholders.	Creation/redemption mechanism generally makes ETFs more tax-efficient.

An illustrative comparison: VTI vs VTSAX.

Python data and implementation patterns

```
# yfinance
```

```
pip install yfinance

import yfinance as yf

ticker = ['SPY']

df = yf.download(ticker, start='2002-1-1')

df = df['close'].asfreq(freq='ME', method='ffill')

R = df.pct_change()

# Tiingo

pip install tiingo

from tiingo import TiingoClient

config = {'api_key': YOUR_API_STRING}

client = TiingoClient(config)

df = client.get_dataframe('SPY', startDate='2002-01-01', frequency='monthly')

R = df['adjClose'].pct_change()
```

The histogram and cumulative-return charts compare equity and bond-like ETFs; cumulative log returns are displayed on a logarithmic scale. The "Mystery Fund" chart is a reminder that an unusually smooth return history deserves scrutiny rather than automatic admiration.

16.2 Estimation error: why theoretical optimal weights can fail

To estimate the mean–variance frontier you need N expected returns, N variances, and N(N−1)/2 covariances. Historical estimates are noisy.

- Small input changes can create extremely different "optimal" weights even while forecast portfolio mean/volatility barely change.
- Errors are magnified: garbage in, garbage out. Michaud (1989) calls mean–variance optimizers "error-maximizing" portfolios.
- Short moving windows can be procyclical: recent high returns may coincide with high valuations and low future expected returns; apparently calm historical volatility can understate latent/future risk.

Practical robustness measures

- Use an economic expected-return model such as CAPM.
- Impose constraints: w_i≥0 (no shorting), w_i≤c (position caps), turnover limits, max drawdown, tracking-error limits.
- Use robust/simple allocations: 1/N, market-value weights, inverse-volatility/risk parity, minimum variance.

Equal weight

$$w_i = \frac{1}{N}$$

Value weight

$$w_i = \frac{ME_i}{\sum_j ME_j}$$

Simplified risk parity / inverse vol

$$w_i = \frac{\frac{1}{\sigma_i}}{\sum_j \frac{1}{\sigma_j}}$$

Naive diversification — DeMiguel, Garlappi & Uppal (2007)

Across 14 optimized models and 7 empirical datasets, none consistently outperformed 1/N by Sharpe ratio. Estimation error offset the theoretical gains from optimal diversification. The empirical comparison suggests that a sample-based mean–variance strategy might need roughly 250 years of data for 25 assets and ~500 years for 50 assets to beat 1/N reliably.

Strategy	Assumption on means	Volatility	Correlations	Interpretation
Optimal mean-var.	Unconstrained	Unconstrained	Unconstrained	Most complex strategy.
Minimum variance	No mean estimate	Unconstrained	Unconstrained	No need to estimate means.
Risk parity	No mean estimate	Unconstrained	Equal to 0	No need to estimate means/correlations.
Equal weighted	None	None	None	Most simple; nothing to estimate.
Value weighted	Observable market values	Observable/ implicit	Observable/ implicit	Observable.

PART 17

Portfolio Implementation Constraints

Portfolio weights are not executed in a frictionless vacuum. Costs, taxes where applicable, market impact, financing terms, short-sale availability, institutional mandates, and realized drawdowns can all make nominally efficient allocations infeasible. A constraint can be represented in the feasible set or as a penalty in the objective. Such frictions also alter the meaning of backtested performance.

For drifted pretrade weights w^- and target weights w^+, a common one-way turnover convention is $\frac{1}{2}\sum_i |w_i^+ - w_i^-|$, while other conventions omit the factor $1/2$. Drawdown compares current wealth to its previously attained high-water mark and therefore describes path dependence absent from single-period variance.

17.1 Portfolio constraints: turnover and max drawdown

Turnover

Before rebalancing at t, last period's weights drift with realized returns. Observation turnover is the sum of absolute weight changes:

$$\text{Drifted weight}_i = \frac{w_{i,t-1}(1+R_{i,t})}{\sum_j w_{j,t-1}(1+R_{j,t})}$$

$$T_t = \sum_i |w_{i,t} - \text{drifted weight}_i|; T_t \le c$$

• Benefits listed: reduce transaction costs, prevent excessive trading, create portfolio persistence, improve OOS performance.

Worked turnover example: Start w_0=(0.5,0.5); returns (15%,−5%). Drifted weights become 0.575/(0.575+0.475)=0.547619 and 0.452381. Rebalance to (0.5,0.5): turnover = |−0.047619|+|+0.047619|=0.095238 = 9.52% under the two-sided turnover convention.

Maximum drawdown

$$HWM_t = max_{s \le t} V_s$$

$$DD_t = \frac{HWM_t - V_t}{HWM_t}; \quad DD_t \le c$$

• Benefits listed: limit downside exposure, preserve capital in stress, improve risk-adjusted returns, appeal to loss-averse investors.

Worked max-drawdown example: NAV = [100,110,120,110,105,115]. HWM = [100,110,120,120,120,120]. Drawdowns = [0,0,0,8.33%,12.50%,4.17%]. Maximum drawdown = 12.50%.

PART 18

Active Management and Performance Evaluation

A good absolute portfolio and a good active portfolio are evaluated differently. Sharpe ratio measures premium per unit of total risk. Information ratio measures expected benchmark-relative outperformance per unit of *active* risk. A portfolio can have a high Sharpe ratio while taking large unintended benchmark exposures, or a high information ratio while having modest standalone Sharpe.

$$\mathrm{SR} = \frac{E[R_p] - R_f}{\sigma_p}, \qquad \mathrm{IR} = \frac{E[R_p - R_b]}{\sqrt{(w_p - w_b)^\top \Sigma (w_p - w_b)}}.$$

Factor attribution further separates market compensation, systematic style exposures, and unexplained regression residuals. A manager's observed outperformance is a description; attributing that outperformance to skill requires an appropriate model and credible evidence.

18.1 Active portfolio management, tracking error, information ratio

Active return

$$R_a = R_p - R_b$$

Active weight

$$w_{a,i} = w_{p,i} - w_{b,i}; \sum_i w_{a,i} = 0$$

$$R_a = \sum_i w_{a,i} R_i$$

- Positive active weight = overweight vs benchmark; negative = underweight.

Tracking error / active risk

$$\sigma_a = \sqrt{V(R_a)} = \sqrt{(w_a^\top \Sigma w_a)}$$

Information ratio

$$\mathrm{IR} = \frac{\mathbb{E}[R_a]}{\sigma_a}$$

IR range	Observation interpretation
0.25–0.50	Decent performance.
0.50–0.75	Above average / solid active management.
>0.75	Excellent / top-quartile managers.

Worked information-ratio example

Two uncorrelated assets: expected returns 15% and 10%; volatilities 25% and 20%; benchmark weights (0.5,0.5); portfolio weights (0.8,0.2).

$$w_a = (+0.3, -0.3); \mathbb{E}[R_a] = 0.3 \times 0.15 - 0.3 \times 0.10 = 0.015$$
$$V(R_a) = 0.3^2(0.25^2 + 0.20^2) = 0.009225; \sigma_a \approx 0.09605$$
$$\text{IR} \approx \frac{0.015}{0.09605} \approx 0.156$$

18.2 Buffett's Alpha

Frazzini, Kabiller & Pedersen (2018): Berkshire Hathaway realized a Sharpe ratio of 0.79, higher than any other stock or mutual fund with >30 years of history in their study. Berkshire has significant CAPM alpha, but much less/insignificant alpha after controlling for additional risk factors. The paper's explanation: leverage plus a focus on cheap, safe, high-quality stocks.

Berkshire CAPM regression (554 observations)

Term	Coef.	p-value
α	0.0080	0.001
Mkt−RF	0.6582	<0.001

R^2=0.222. CAPM alpha is statistically significant.

Berkshire five-factor regression

Term	Coef.	p-value	Signal
α	0.0045	0.050	Borderline at 5%; much smaller than CAPM alpha.
Mkt−RF	0.8169	<0.001	Market exposure.
SMB	−0.2227	0.008	Large-cap tilt.
HML	0.3980	<0.001	Value exposure.
RMW	0.4205	<0.001	Quality/profitability exposure.
CMA	0.0541	0.715	No significant investment-factor exposure here.

R^2=0.322. Additional factors explain more variation and absorb much of the abnormal return.

PART 19

Market Efficiency, Quantitative Research, and Limits of Prediction

Even logically coherent strategies may fail in live markets. Security prices quickly incorporate public information, investment opportunities attract competing capital, and backtests can overstate significance when many specifications are tried. In finance, *prediction* and *causal explanation* are valuable but different goals.

The Grossman–Stiglitz argument highlights a paradox of perfectly efficient prices: if information were reflected in prices without compensation for research, no one would pay to acquire it. Markets can be close to efficient precisely because informed participants compete, while limited inefficiencies persist where information acquisition, arbitrage risk, or transaction costs matter.

$$\boxed{\begin{array}{c}\text{economic mechanism} + \text{statistics} + \text{out-of-sample validation} \\ + \text{costs} + \text{competition} + \text{implementation}\end{array}}.$$

The result is not a single guaranteed signal but a disciplined process that connects economic value, statistical estimation, portfolio optimization, implementation, and measurement.

19.1 Machine learning: two cultures and finance-specific difficulties

Culture	Priority
Structural model culture (economist)	Impose fully/partly specified economic structure; study mechanisms; test hypotheses.
Prediction model culture (data scientist)	Prioritize statistical explanatory/predictive power; less emphasis on structural mechanism.

General AI examples in common financial applications: DeepMind, ChatGPT, Amazon/Netflix recommender systems, self-driving cars, image recognition. A cat-vs-dog image is used as a standard classification analogy.

Why ML is unusually hard in finance: Expected returns are not directly observed; macro-finance often has only a few hundred monthly observations; new data arrive only through time; signal-to-noise is weak because profit-seeking competition erodes predictability; prices react to unanticipated news; investors learn; regulation/technology/economic structure change, making the data-generating process unstable.

19.2 Efficient markets, near-efficient markets, and random walks

Fama (1970) Efficient Market Hypothesis (EMH): security prices fully reflect available information.

Form	Information already reflected in price
Weak	All information in the history of stock trading.
Semi-strong	All publicly available information.
Strong	All relevant information, including inside information.

• Investors will spend resources to analyze new information as long as the activity generates sufficiently higher returns. Regulation of congressional securities trading also changed with the 2012 STOCK Act.

Grossman–Stiglitz near-efficiency logic

1. Suppose gathering information and trading on it are costly.

2. If prices already reflected every piece of public information perfectly, nobody would pay to gather information.

3. If nobody gathered information, that information could not get into prices.

4. Therefore a realistic equilibrium is near-efficient: competition makes markets almost efficient, but some inefficiency must remain to compensate informed investors for costs and risks. Limits to arbitrage also matter.

View	Implication for active management
Efficient Market Hypothesis	If all relevant information is in prices, attempts to beat the market are futile; active investors paying fees should underperform by roughly the fee.
Inefficient Market Hypothesis	If prices are strongly driven by irrationality/bias, beating the market should be relatively easy.
Near-Efficient Market Hypothesis	Professional competition makes markets almost efficient, yet some profitable opportunities remain; only a limited amount of capital can exploit them before they disappear.

Random walk theory: stock prices evolve in a largely unpredictable way because available information is already embedded in prices; future movements are independent of past movements. Random price changes are therefore not evidence of irrationality. Black–Scholes and binomial option-pricing models use random-walk-style price dynamics.

19.3 Statistical failure modes and fund-manager evidence

• Twitter and Zillow are shown as cautionary data-science failures. The Zillow graphic shows an ML forecast of $500k for two houses, with one seller accepting the offer and the other rejecting it—illustrating that real-world deployment can create selection problems when counterparties respond to a model.

• Notable hedge-fund managers have varied academic backgrounds: George Soros (LSE MS), David Tepper (CMU MBA), David Booth (Chicago MBA), Cliff Asness (Chicago PhD Finance), Ken Griffin (Harvard BA Economics), James Simons (Berkeley PhD Math), Ray Dalio (Harvard MBA), Israel Englander (NYU BS Finance), John Campbell (Yale PhD Economics), David Shaw (Stanford PhD CS), Dmitry Balyasny (Loyola Chicago BBA), Steven A. Cohen (Penn BS Economics).

19.4 Empirical evidence on machine learning in asset markets

Paper	Main analysis takeaway
Lopez-Lira & Tang (working paper, 2024)	LLMs such as ChatGPT can extract sentiment from headlines that predicts out-of-sample daily returns; predictability is stronger for smaller stocks and after negative news.
Chinco, Clark-Joseph & Ye, JF 2019	LASSO on the cross-section of lagged returns improves one-minute-ahead OOS fit and forecast-implied Sharpe ratios. Successful predictors are sparse, short-lived, unexpected, and often tied to fundamental news.
Cohen, Malloy & Nguyen, JF 2020, "Lazy Prices"	Changes in 10-K/10-Q language/construction predict future returns and operations. Short "changers" / long "non-changers" earns up to 188 bp/month alpha; lack of announcement effect points to investor inattention.
Jiang, Kelly & Xiu, JF 2023	Image-analysis ML on price charts finds predictive trend patterns beyond predefined momentum/reversal, robust across horizons and international markets.
Cao, Jiang, Wang & Yang, JFE 2024	AI analysts can outperform most humans when information is transparent but voluminous; humans retain an edge when institutional knowledge/intangibles/distress matter. Human + machine reduces extreme errors.

KEY: The central finding is not "markets are either predictable or unpredictable." Competition makes easy signals disappear, but costly information, frictions, limits to arbitrage, and new data/algorithms can leave narrow, unstable pockets of predictability.

APPENDIX A

Complete Concept Index

This index organizes the principal ideas by their economic role rather than by any particular teaching sequence. References to the main parts follow the investment decision process.

A.1 Uncertainty and measurement

- Finite state space.
- State-contingent payoffs.
- Risk-free versus risky returns.
- Realized and conditional expected returns.
- Surprises.
- Simple and log returns.
- Total return.
- Distributional moments.
- Sample moments.
- Arithmetic/geometric growth.
- Equity and share adjustments.

A.2 Economic valuation

- Present value.
- Replication.
- Arbitrage.
- Law of one price.
- Complete markets.
- Arrow–debreu claims.
- State-price matrix.
- Stochastic discount factor.
- Consumption utility.
- Jensen.
- Certainty equivalents.
- Risk premia.
- Covariance with bad states.

A.3 Derivative valuation

- Physical and risk-neutral probabilities.
- Likelihood ratios.
- Martingale pricing.
- Monte carlo.
- Binomial call.
- Black–scholes.
- Girsanov.
- d_1, d_2.
- Delta hedge.
- Numeraire.
- Fundamental theorem of asset pricing.

A.4 Conditional expected returns

- Dividend discounting.
- Gordon growth.
- Payout retention and roe.
- Dividend and earnings yields.
- Macro predictors.
- Ols significance.
- Confidence, type i errors.
- Look-ahead bias.
- Rolling/expanding estimation.
- Out-of-sample r^2.
- Style signals and factor sorts.

A.5 Joint and conditional risk

- Covariance.
- Correlation.
- Two-asset portfolios.
- Diversification and concentration.
- Systematic/idiosyncratic shocks.
- Factor risk models.
- Capm risk decomposition.
- Barra style risks.
- Ewma.
- Arch.
- Garch.

- Heston.
- Heston–nandi.
- Volatility clustering.

A.6 Static portfolio construction

- Markowitz.
- Feasible set.
- Global minimum variance.
- Efficient frontier.
- Short-sale constraints.
- a, b, c, d.
- Capital allocation lines.
- Tangency risky composition w_t.
- Sharpe ratios.
- Two-fund separation.
- Indifference curves.
- Risky exposure v.
- Final weights $x = vw_t$.

A.7 Dynamic portfolio allocation

- Return timing.
- Volatility timing.
- The conditional opportunity set.
- Fixed and variable risk aversion.
- Random-walk discipline.
- Missed-best-days evidence.
- Perfect-timer benchmark.
- Treynor–mazuy.
- Signal timing constraints.

A.8 Equilibrium and testing

- Market clearing.
- Value-weighted market portfolio.
- Capm.
- Cml versus sml.
- Beta versus volatility.
- Residual alpha.
- Explanatory r^2.

- Macro factor prices.
- Fama–french smb/hml.
- Rmw/cma/mom.
- Alpha significance.

A.9 Trading and robustness

- Long–short investing.
- Dollar versus beta neutrality.
- Liquidity and leverage.
- Naive $1/n$.
- Shrinkage.
- Risk parity.
- Minimum variance.
- Black–litterman.
- Hierarchical risk parity.
- Optimization error.
- Turnover.
- Market impact.
- Drawdown.
- Transaction costs.

A.10 Evaluation and market structure

- Total versus active risk.
- Tracking error.
- Information ratio.
- Regression attribution.
- Buffett factors.
- Emh.
- Grossman–stiglitz information incentives.
- Low signal-to-noise.
- Structural instability.
- Data mining.
- Quantitative research discipline.

APPENDIX B

Complete Formula Index

This reference groups formulas by the question they answer. The formulas assume internally consistent units and horizons (simple versus log, daily versus annual). When gross returns are used, $1+R$ is distinguished from the simple net return R.

B.1 Returns, compounding, and distributional moments

$$R_t^x = \frac{P_t}{P_{t-1}} - 1$$

$$R_t = \frac{P_t + D_t}{P_{t-1}} - 1$$

$$R_{t\to T} = \prod (1 + R_s) - 1$$

$$r_t = \ln(1 + R_t), \text{cumulative log return} = \sum r_t$$

$$\bar{R} = \frac{1}{T}\sum R_t\,; \quad \bar{R}_G = \left[\prod (1 + R_t)\right]^{\frac{1}{T}} - 1$$

$$V(R) = \mathbb{E}[(R - \mathbb{E}[R])^2], \sigma = \sqrt{V}$$

$$V_d = \mathbb{E}[\min(R - \tau, 0)^2], \sigma_d = \sqrt{V_d}$$

$$\text{Skew} = \frac{\mathbb{E}[(R - \mathbb{E}[R])^3]}{\sigma^3},$$

$$\text{Kurt} = \frac{\mathbb{E}[(R - \mathbb{E}[R])^4]}{\sigma^4},$$

$$\text{excess kurtosis} = \text{Kurt} - 3$$

B.2 Cash-flow valuation and predictive regressions

$$P_0 = \frac{X_T}{(1+R)^T}$$

$$P_t = \frac{\mathbb{E}_t[D_{t+1}]}{R - G}$$

$$R = \frac{\mathbb{E}_t[D_{t+1}]}{P_t} + G$$

$$R_m^e = R_m - R_f$$

$$R_{m,t}^e = a + bF_{t-1} + \varepsilon_t$$

$$\hat{\beta}_1 = \frac{\mathrm{Cov}(x, y)}{\mathrm{Var}(x)}\,; \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$t = \frac{\hat{\beta}_1}{\mathrm{SE}(\hat{\beta}_1)}$$

B.3 State prices, stochastic discount factors, and risk-neutral pricing

$$P(X) = \sum \zeta(\omega) X(\omega)$$

$$M(\omega) = \frac{\zeta(\omega)}{P(\omega)}$$

$$P(X) = \mathbb{E}[MX]$$

$$\mathbb{E}[M] = \frac{1}{1 + R_f}$$

$$1 = \mathbb{E}[M(1 + R)]$$

$$\mathbb{E}[1 + R] = (1 + R_f)(1 - \mathrm{Cov}(M, R))$$

$$\mathbb{E}[R^e] = -(1 + R_f)\,\mathrm{Cov}(M, R^e)$$

$$\frac{Q}{P} = \frac{M}{\mathbb{E}[M]}$$

$$P(X) = \frac{\mathbb{E}_Q[X]}{1 + R_f}$$

$$\mathbb{E}_Q[R] = R_f$$

B.4 Black–Scholes pricing and delta

$$C_0 = S_0\Phi(d_1) - Ke^{-r_f T}\Phi(d_2)$$

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + (r_f + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

$$\Delta = \Phi(d_1)$$

B.5 Static and dynamic asset allocation

$$U = \mathbb{E}[R] - \frac{\gamma}{2}V(R)$$

$$R_p = R_f + \upsilon R_m^e$$

CAL

$$\mathbb{E}[R_p] = R_f + \frac{\mathbb{E}(R_m^e)}{\sigma_m}\sigma_p$$

$$\upsilon^* = \frac{\mathbb{E}[R_m^e]}{\gamma V_m}$$

$$\upsilon_t^* = \frac{\mathbb{E}_t[R_{m,t+1}^e]}{\gamma V_t[R_{m,t+1}]}$$

Treynor-Mazuy

$$R_p^e = \alpha + \beta_1 R_m^e + \beta_2 (R_m^e)^2 + \varepsilon,$$

$$\beta_2 > 0 \;\rightarrow\; \text{timing evidence}$$

$$\text{OOS } R^2 = 1 - \frac{\text{SSE}_{\text{model}}}{\text{SSE}_{\text{historical_mean}}}$$

B.6 Volatility and conditional variance models

EWMA (analysis recursion)

$$\sigma_t^2 = \alpha R_{t-1}^2 + (1-\alpha)\sigma_{t-1}^2$$

GARCH(1,1)

$$\sigma_t^2 = k + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

Heston

$$dS_t = \mu S_t dt + \sqrt{\nu}_t S_t dW_t^S; d\nu_t = \kappa(\theta - \nu_t)dt + \sigma\sqrt{\nu}_t dW_t^\nu$$

Heston–Nandi

$$\sigma_t^2 = \omega + \beta\sigma_{t-1}^2 + \alpha(\varepsilon_{t-1} - \gamma\sigma_{t-1})^2$$

B.7 Mean–variance portfolio optimization

$$\mathbb{E}[R_p] = w^\top \mu; V(R_p) = w^\top \Sigma w$$

$$w_{\text{MVP}} = \frac{\Sigma^{-1}1}{1^\top \Sigma^{-1}1}$$

$$w_{\text{SR}} = \frac{\Sigma^{-1}(\mu - 1R_f)}{1^\top \Sigma^{-1}(\mu - 1R_f)}$$

$$\text{SR} = \frac{\mathbb{E}[R_p] - R_f}{\sigma_p}$$

$$\text{Tracking error} = \sqrt{w_a^\top \Sigma w_a}$$

$$\text{IR} = \frac{\mathbb{E}[R_a]}{\text{tracking error}}$$

Equal-weight diversification

$$V(R_p) = \frac{1}{N}\bar{\sigma}^2 + \frac{N-1}{N}\overline{\text{Cov}}$$

B.8 CAPM and multifactor expected returns

CAPM

$$\mathbb{E}[R_i] - R_f = \beta_i(\mathbb{E}[R_m] - R_f)$$

$$\beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{V(R_m)}$$

CAPM test

$$R_i^e = \alpha_i + \beta_i R_m^e + \varepsilon_i$$

CAPM risk decomposition

$$V(R_i^e) = \beta_i^2 V(R_m^e) + V(\varepsilon_i)$$

One-factor risk-premium notation

$$\mathbb{E}[R_i^e] = \beta_i \lambda$$

Fama–French 3

$$R_i^e = \alpha + \beta_m MKT + \beta_{\mathrm{SMB}}\mathrm{SMB} + \beta_{\mathrm{HML}}\mathrm{HML} + \varepsilon$$

$$\mathbb{E}[R_i] - R_f = \alpha_i + \beta_i^m(\mathbb{E}[R_m] - R_f) + \beta_i^{SMB}\mathbb{E}[R_{SMB}] + \beta_i^{HML}\mathbb{E}[R_{HML}]$$

Multifactor SDF

$$M = a + \sum b_k F_k$$

Expected premium

$$\mathbb{E}[R_i^e] = \sum \beta_i^k \lambda_k$$

B.9 Additional identities and implementation equations

Equation	Meaning
$R_t^{\text{total}} = \frac{P_t - P_{t-1} + D_t}{P_{t-1}}$	Cash-inclusive simple return
$1 + R_{0:T} = \prod_{t=1}^{T}(1 + R_t)$	Compounded wealth growth
$\text{Cov}(X, Y) = E[(X - EX)(Y - EY)]$	Joint variation
$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$	Scaled comovement
$\sigma_p^2 = w^\top \Sigma w$	Portfolio total variance
$\Sigma = B\Sigma_F B^\top + D$	Factor plus specific covariance
$G = (1 - D/E)ROE$	Retained-earnings growth
$\widehat{\mu}_{i,t}^e = \delta_0 + \sum_j \delta_j z_{j,i,t}$	Characteristic-based alpha
$z_i = (x_i - \bar{x})/s_x$	Standardized characteristic
$\sigma_{\text{GMVP}}^2 = (\mathbf{1}^\top \Sigma^{-1} \mathbf{1})^{-1}$	Global minimum variance
$w_T = \frac{\Sigma^{-1}\mu^e}{\mathbf{1}^\top \Sigma^{-1}\mu^e}$	Tangency composition
$\upsilon^* = \frac{\mu_T^e}{\gamma \sigma_T^2}$	Investor risky scale
$x^* = \gamma^{-1}\Sigma^{-1}\mu^e$	Final risky holdings
$w_{m,i} = ME_i / \sum_j ME_j$	Market-cap weight
$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\sigma_m^2}$	Market-risk exposure
$\lambda_k = -(1 + R_f) b_k \text{Var}(F_k)$	Price per factor risk unit
$R_a = R_p - R_b$	Benchmark-relative return
$\sigma_a = \sqrt{(w_p - w_b)^\top \Sigma (w_p - w_b)}$	Active tracking error
$\text{IR} = E[R_a]/\sigma_a$	Information ratio

Notation and assumptions. $\mu^e = \mu - R_f \mathbf{1}$; w_T is a normalized risky composition, υ a risky wealth fraction, and $x = \upsilon w_T$ the final risky positions. Factor decompositions presume compatible residual orthogonality and covariance assumptions. With correlated factors, risk premia must use appropriate multivariate betas and covariances.

APPENDIX C

Final Investment Discipline Map

The central result is not a particular estimator or portfolio formula but the consistent transfer of information from economic valuation to expected returns, risk, investor preferences, implementable holdings, and ex post evaluation.

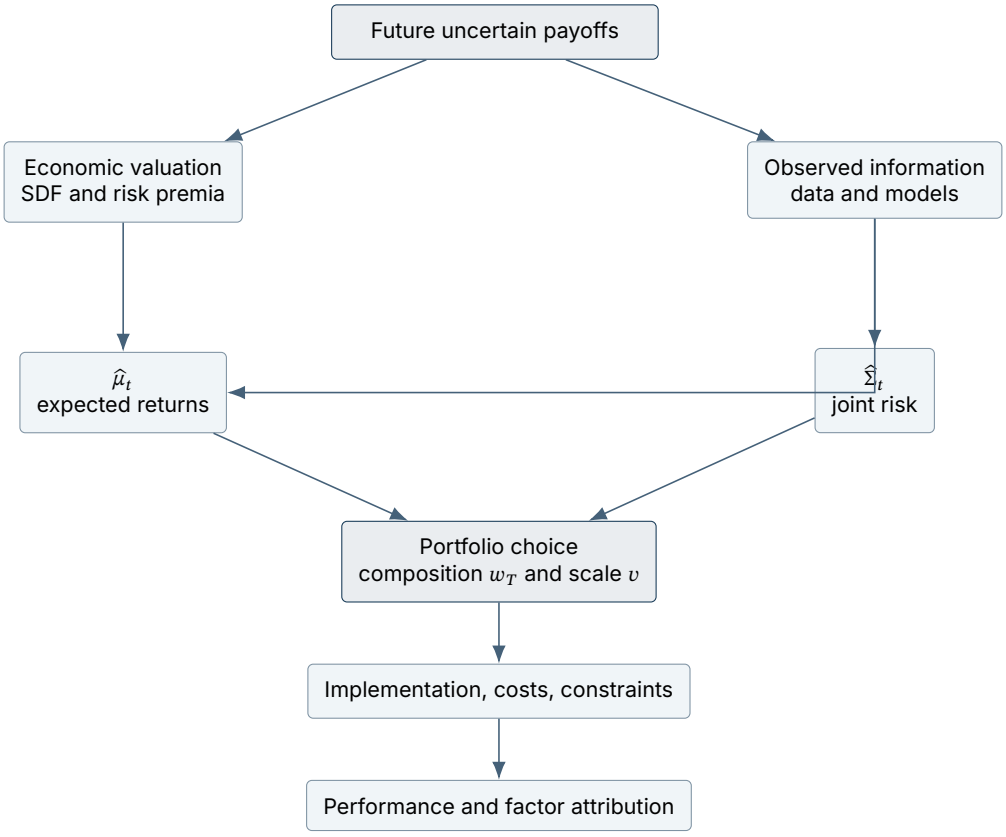


Figure 10: The investment decision as a sequence from economic uncertainty to measurable positions.

C.1 Two connected pricing problems

The SDF maps state payoffs into prices. Its covariance with an asset's payoff explains the sign and magnitude of required risk premia. A change of measure built from the same SDF converts that valuation into a risk-neutral expectation, particularly useful for derivatives. Those pricing restrictions and empirical forecasts inform μ; contemporaneous joint-risk models inform Σ. Neither can replace the other.

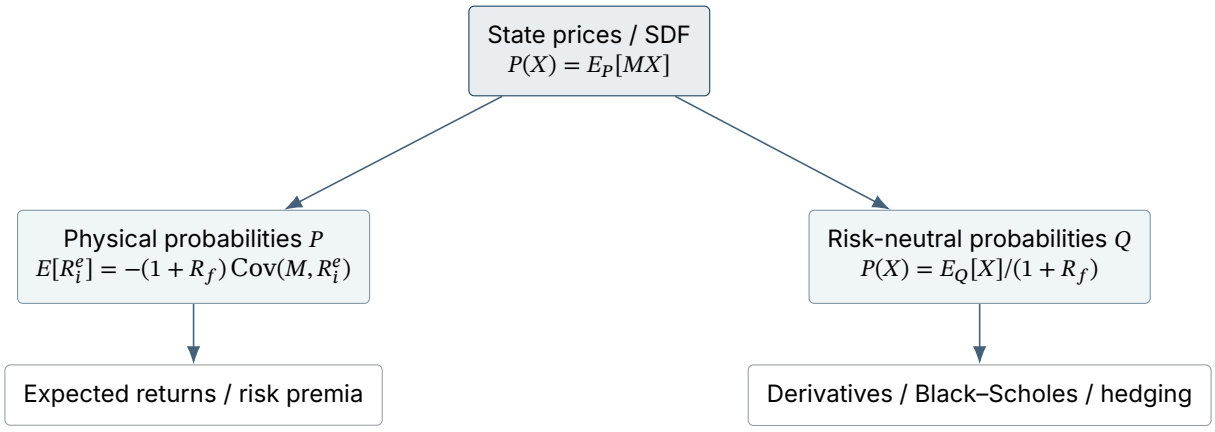


Figure 11: The same SDF supports both equilibrium premia and arbitrage-free derivative pricing.

APPENDIX D

Quantitative Investment Research Workflow

A quantitative investment process distinguishes expected-return research, covariance estimation, portfolio construction, and execution. Each stage should be tested against information availability, live tradability, and a suitable benchmark.

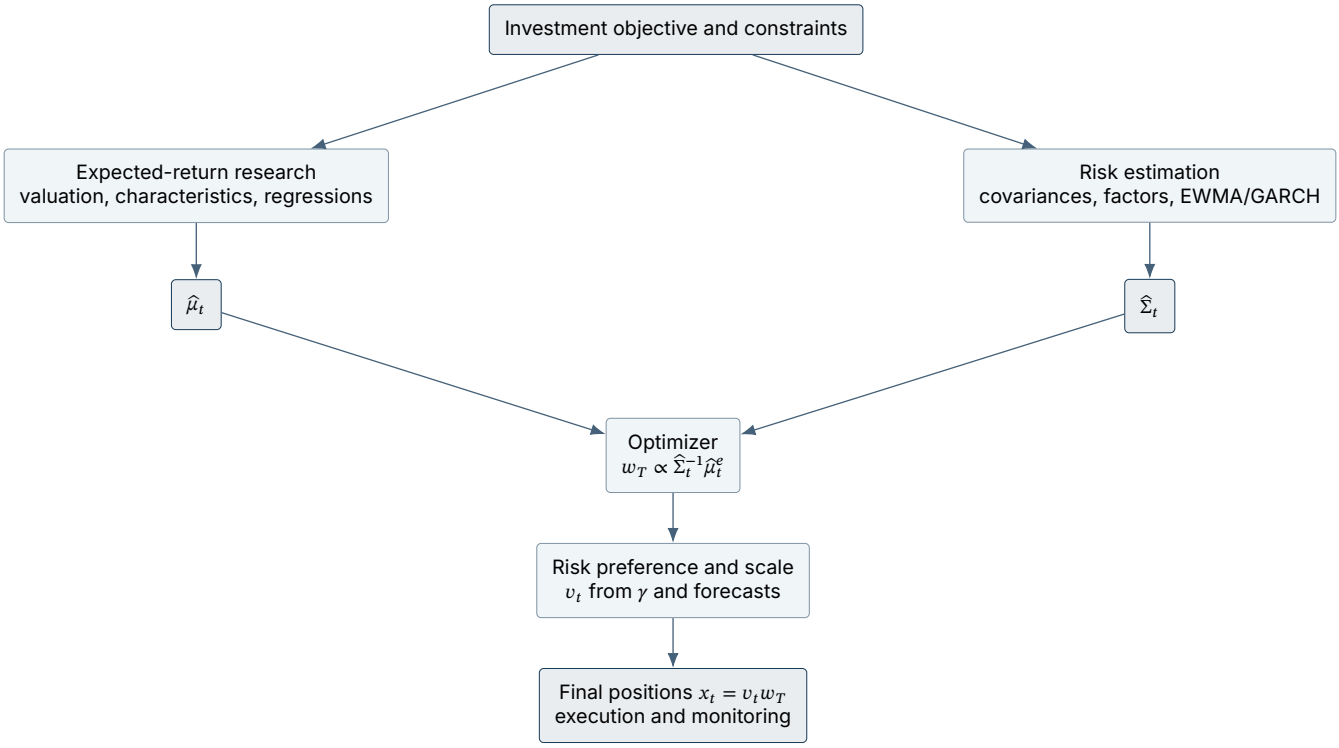


Figure 12: A practical research pipeline from an investment objective to risk-scaled positions.

D.1 Operational interpretation

The forecast vector $\hat{\mu}_t$ should be produced from lagged, reproducible signals and validated out of sample. The estimated $\hat{\Sigma}_t$ captures the contemporaneous covariance structure and should be stress-tested for model error. The optimizer chooses a risky composition subject to realistic mandates; investors or portfolio policies select a total scale consistent with risk tolerance and financing capacity. Orders, turnover, trading costs, and benchmark attribution transform a theoretically optimal allocation into an accountable investment process.

ABOUT THE AUTHOR

Mizuki Seika

Quantitative Trader · Web3 Explorer · Independent Idol & Music Producer

I'm Mizuki Seika, a quantitative trader working at the intersection of mathematics, financial markets, and emerging technology; an explorer of Web3 and cryptocurrency; and, in a parallel universe that happens to be equally real, an independent idol and original music producer.

My academic path spans Carnegie Mellon University's Master of Science in Computational Finance (MSCF '28), Mathematics at Tsinghua University ('25), and an exchange year at the University of Oxford ('24). My interests range from quantitative research and systematic trading to decentralized finance and digital assets.

To me, markets and music have more in common than they appear to: both reward an ear for signals, a sense of structure, and the courage to build something original.

> *In finance, I turn complexity into models.*
> *In music, I turn it into melodies.*

This work is where I bring that same creative discipline to learning --- reconstructing knowledge until the pieces form a framework that actually makes sense.

Explore My Worlds

LinkedIn
Original music

GitHub
Open-source projects and research

Spotify
Original music

Instagram
The creative side, on and off stage

Thank you for reading.